MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

MULTIVIBRATOR, MONOSTABLE, DIGITAL, KEY CODE BDE

INACTIVE FOR NEW DESIGN AFTER DATE OF THIS SPECIFICATION

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic monostable multivibrator module.

1.2 Classification. This specification covers one class of modules.

Class I as specified in MIL-STD-1389.

2. APPLICABLE DOCUMENTS

2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Naval Electronic Systems Command, ATTN: ELEX 8111, Department of the Navy, Washington, DC 20360, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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3.3 Failure rate requirements. All modules acquired to this specification shall have a predicted failure rate no greater than 1.9 failures per million operating hours when predicted in accordance with MIL-STD-1389.

3.4 Performance requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class I as specified in MIL-STD-1389.

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be BDE.

3.5.2 Size. The module size shall be 1A.

3.5.3 Weight. The module shall have a maximum weight of 75 grams.

3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.5.6 Depth The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

* 3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/118-1, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance tests All production modules shall be tested in accordance with footnotes 1 and 2 of table I.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I (see 6.2).

3.9 Power supply voltage. The module shall be capable of conforming to the requirements of table I for a power supply voltage (V_{CC}) of 5.0 ±0.5 Vdc, except that the limits for output short circuit current apply only for a power supply voltage of 5.5 Vdc ±1 percent.

4. QUALITY ASSURANGE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

* 4.3 Test equipment Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

4.4 Test procedures. The module shall be tested as specified in table I.

4.4.1 Alternate test methods Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity and they conform to the requirements of MIL-M-28787.

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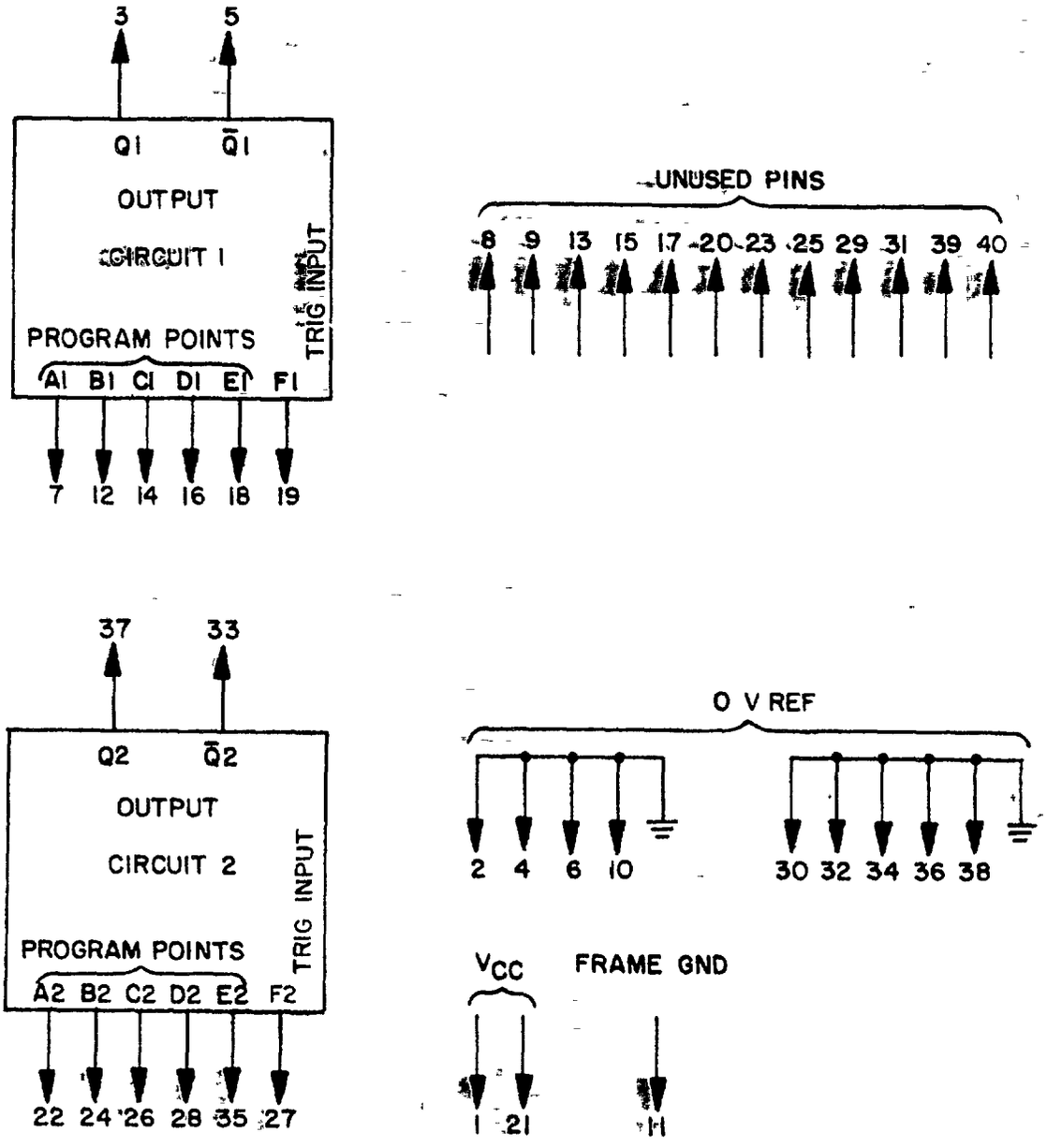


FIGURE 1. BDE module function diagram, monostable multivibrator.

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* TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/ 2/	Acceptance requirements 25°±5°C ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
A1-A7	Continuity	II	X	0	1	0	1	0	1	0	1	Ω
A8	Continuity	II	S	0	1	0	1	0	1	0	1	Ω
B1,B2	Isolation	II	X	10	∞	10	∞	10	∞	10	∞	MΩ
C1,C5	Output, low level	III	X	0.0	0.4	0.0	0.45	0.0	0.4	0.0	0.45	V
C3,C7	Output, high level	III	X	3.8	5.5	3.5	5.5	3.8	5.5	3.5	5.5	V
C4,C8	Output, low level	III	X	0.0	0.4	0.0	0.45	0.0	0.4	0.0	0.45	V
C2,C6	Output, high level	III	X	3.8	5.5	3.5	5.5	3.8	5.5	3.5	5.5	V
D1-D4	Output short circuit current	IV	X	16	65	16	65	16	65	16	65	mA
D5,D7	Input current, high level	IV	X	0.0	4.0	0.0	8.0	0.0	6.0	0.0	10.0	μA
D6,D8	Input current, low level	IV	X	0.0	3.2	0.0	3.2	0.0	3.2	0.0	3.2	mA
E1,E2	Power supply current	V	S	0	120	0	120	0	120	0	120	mA
F1-F4	Output pulse width	VI	X	9.3	10.7	9.1	10.9	9.0	11.0	8.8	11.2	μs
F5-F8	Output pulse width	VI	X	18.6	21.4	18.2	21.8	18.0	22.0	17.6	22.4	μs
F9-F12	Output pulse width	VI	S	27.9	32.1	27.3	32.7	27.0	33.0	26.4	33.6	μs
F13-F16	Output pulse width	VI	X	37.2	42.8	36.4	43.6	36.0	44.0	35.2	44.8	μs
F17-F20	Output pulse width	VI	S	46.5	53.5	45.5	54.5	45.0	55.0	44.0	56.0	μs
F21-F24	Output pulse width	VI	S	55.8	64.2	54.6	65.4	54.0	66.0	52.8	67.2	μs
F25-F28	Output pulse width	VI	S	65.1	74.9	63.7	76.3	63.0	77.0	61.6	78.4	μs
F29-F32	Output pulse width	VI	X	74.4	85.6	72.8	87.2	72.0	88.0	70.4	89.6	μs
F33-F36	Output pulse width	VI	S	83.7	96.3	81.9	98.1	81.0	99.0	79.2	100.8	μs
F37-F40	Output pulse width	VI	S	93	107	91	109	90	110	88	112	μs

See footnotes at end of table.

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* TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/,2/	Acceptance requirements 25°±5°C ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
F41-F44	Output pulse width	VI	S	102	118	100	120	99	121	96.8	123.0	µs
F45-F48	Output pulse width	VI	S	112	128	109	131	108	132	106	134	µs
F49-F52	Output pulse width	VI	S	121	139	118	142	117	143	114	146	µs
F53-F56	Output pulse width	VI	S	130	150	127	153	126	154	123	157	µs
F57-F60	Output pulse width	VI	X	139	161	136	164	135	165	132	168	µs
G1-G5	Transition time (t_{THL})	VII	S	3	40	3	45	3	40	3	45	ns
G6-G10	Transition time (t_{TLH})	VII	S	10	250	10	250	10	250	10	250	ns
H1,H2,H4	Propagation delay time (t_{PLH})	VIII	S	10	170	10	170	10	170	10	170	ns
H3,H5,H6	Propagation delay time (t_{PHL})	VIII	S	7	170	7	170	7	170	7	170	ns
J1-J4	Output pulse amplitude, low level	IX	S	0.0	0.4	0.00	0.45	0.0	0.4	0.00	0.45	V
J5-J8	Output pulse amplitude, high level	IX	S	3.8	5.5	3.5	5.5	3.8	5.5	3.5	5.5	V
J9-J12	Overshoot	IX	S	0.0	0.5	0.0	0.5	0.0	0.5	0.0	0.5	V
J13-J16	Undershoot	IX	S	0.0	0.5	0.0	0.5	0.0	0.5	0.0	0.5	V
J17-J20	Preshoot	IX	S	0.0	0.5	0.0	0.5	0.0	0.5	0.0	0.5	V
K1,K3	Pulse width (recovery time)	X	S	9.3	10.7	9.1	10.9	9.0	11.0	8.8	11.2	µs
K2,K4	Pulse width (recovery time)	X	S	139	161	136	164	135	165	132	168	µs
L1-L4	Input trigger sensitivity	XI	S	9.3	10.7	9.1	10.9	9.0	11.0	8.8	11.2	µs
L5-L8	Input immunity (20 ns)	XI	S	0	5	0	5	0	5	0	5	ns
L9-L12	Input immunity (0.95 V)	XI	S	0	5	0	5	0	5	0	5	ns

See footnotes at end of table.

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* TABLE I. *Module electrical requirements - Continued.*

Test condition	Parameter	Ref test table	Test type 1/,2/	Acceptance requirements 25°±5°C ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
M1,M2	Power supply, transient rise time	XII	S	9	53	9	53	9	53	9	53	µs
M3,M4	Power supply, transient rise time	XII	S	1.5	40.0	1.5	40.0	1.5	40 0	1.5	40.0	µs
M5-M8	Power supply, false trigger	XII	S	2.0	5.5	2.0	5.5	2.0	5.5	2.0	5.5	V
N1-N12	Crosstalk	XIII	S	0	5	0	5	0	5	0	5	ns
P1,P2	Zener applied potential	XIV	S	3.8	5 5	3.5	5.5	3.8	5.5	3.5	5.5	V
P3,P4	Trigger voltage high	XIV	S	0.0	0.4	0.00	0.45	0.0	0.4	0.00	0.45	V
Q1,Q2	V_{OUT} (low loaded)	XV	S	0.0	1.1	0.0	1.1	0.0	1.1	0.0	1.1	V

1/ "X" parameters shall be 100 percent tested during module acceptance tests.

2/ "S" parameters shall be sample acceptance tested (one percent or two modules from each inspection lot, whichever is greater) at 25°C.

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4.4.2 Order of testing. Test conditions A through Q shall be performed in any order with the following exceptions. If test conditions P or Q are part of a sequence of tests, the order of testing shall be as follows: P, Q where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through Q in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25° ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions A, C through F16, F29 through F36, F53 through F56, G5, G7, H2, H6, L1 through L4, and M shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 3.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 4. Following the vibration tests, the electrical tests of table I which are marked with an "X" in the test type column shall be performed using the initial limits at 25° ±5°C ambient.

4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3. Following the life test and while still at the maximum class temperature, electrical tests A through Q shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table I. Following the return of the module to 25°C, electrical tests A through N shall be performed in the order specified in 4.4.2 using the 25° ±5°C ambient end-of-life requirements of table I.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life (see 3.8).

* 6.3 Cross reference of classification. Military Specification MIL-M-28787/118 supersedes Department of the Navy, Naval Ordnance Systems Command Drawing No. 10001-2658638. Part number M28787/118-1 supersedes part numbers 2658638 and M28787/118.

6.4 Module description. The following electrical parameters reflect the 0° to 60°C end-of-life limits from table I.

Supply voltage	5.0 Vdc ±10%
Maximum power dissipation	660 mW (outputs not loaded)
Input voltage	
Low level	0.0 V minimum, 0.95 V maximum
High level	3.2 V minimum, V_{CC} maximum

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Input current	
Low level	-3.2 mA maximum
High level	10.0 µA maximum
Output voltage	
Low level	0.0 V minimum, 0.45 V maximum
High level	3.5 V minimum, 5.5 V maximum
Output drive current maximum capability	
Low level (V_{OL} = 0.4 V)	35.2 mA
High level (V_{OH} = 3.8 V)	-200 µA
Pulse amplitude	
Low level (V_{OL} = 0.4 V)	0.0 V minimum, 0.45 V maximum
High level (V_{OH} = 3.8 V)	3.8 V minimum, 5.5 V maximum
Input trigger pulse width	85 ns minimum
Output pulse width range	10.0 µs minimum, 150 µs maximum (in 10 µs increments)
Propagation delay time	7.0 ns minimum, 170 ns maximum
Recovery time	26 µs maximum

6.5 Module function. This module contains two monostable multivibrator circuits which have output pulse widths programmable from 10 to 150 µs in 10 µs increments.

6.5.1 Parallel outputs Paralleling of outputs is not permissible.

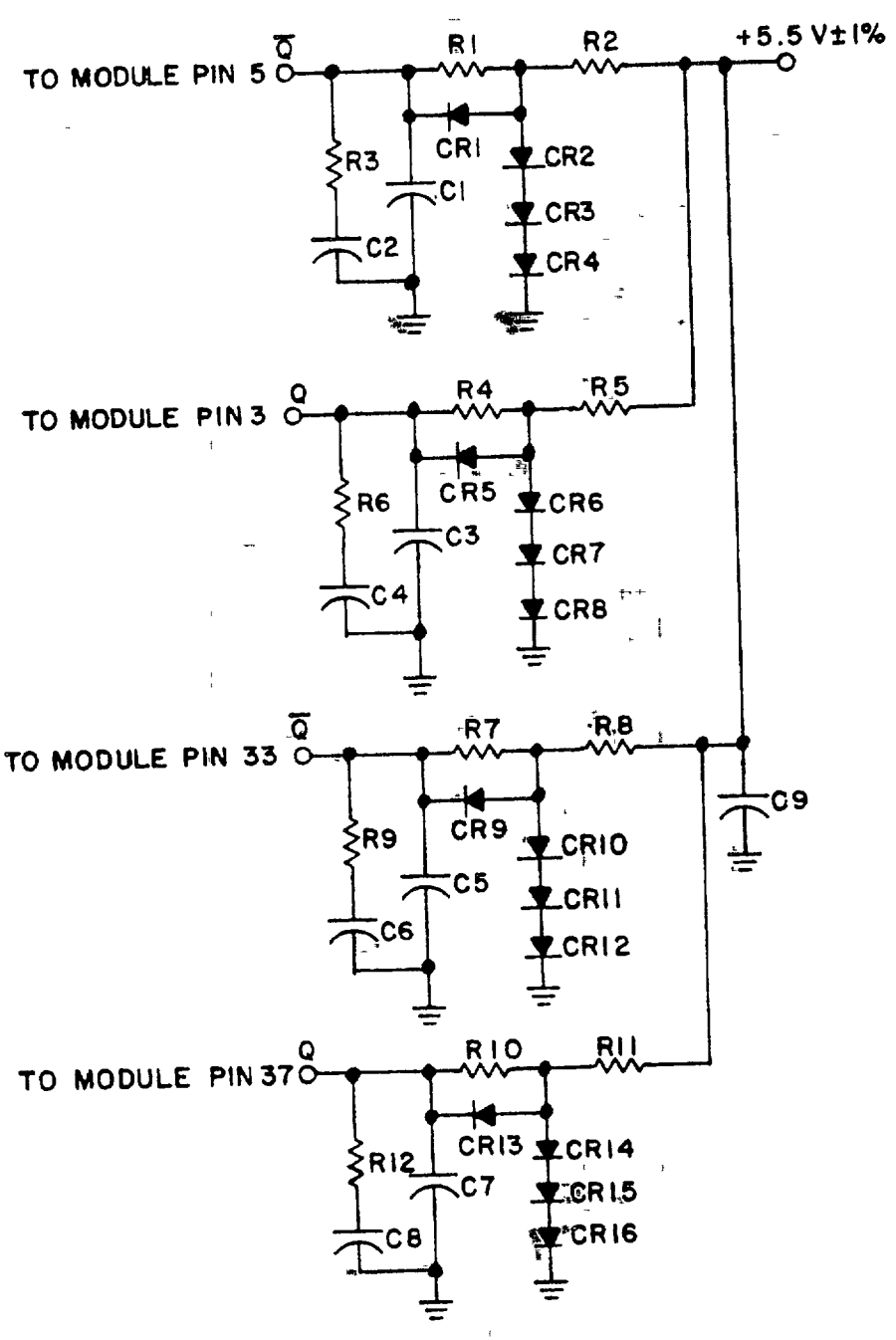
6.6 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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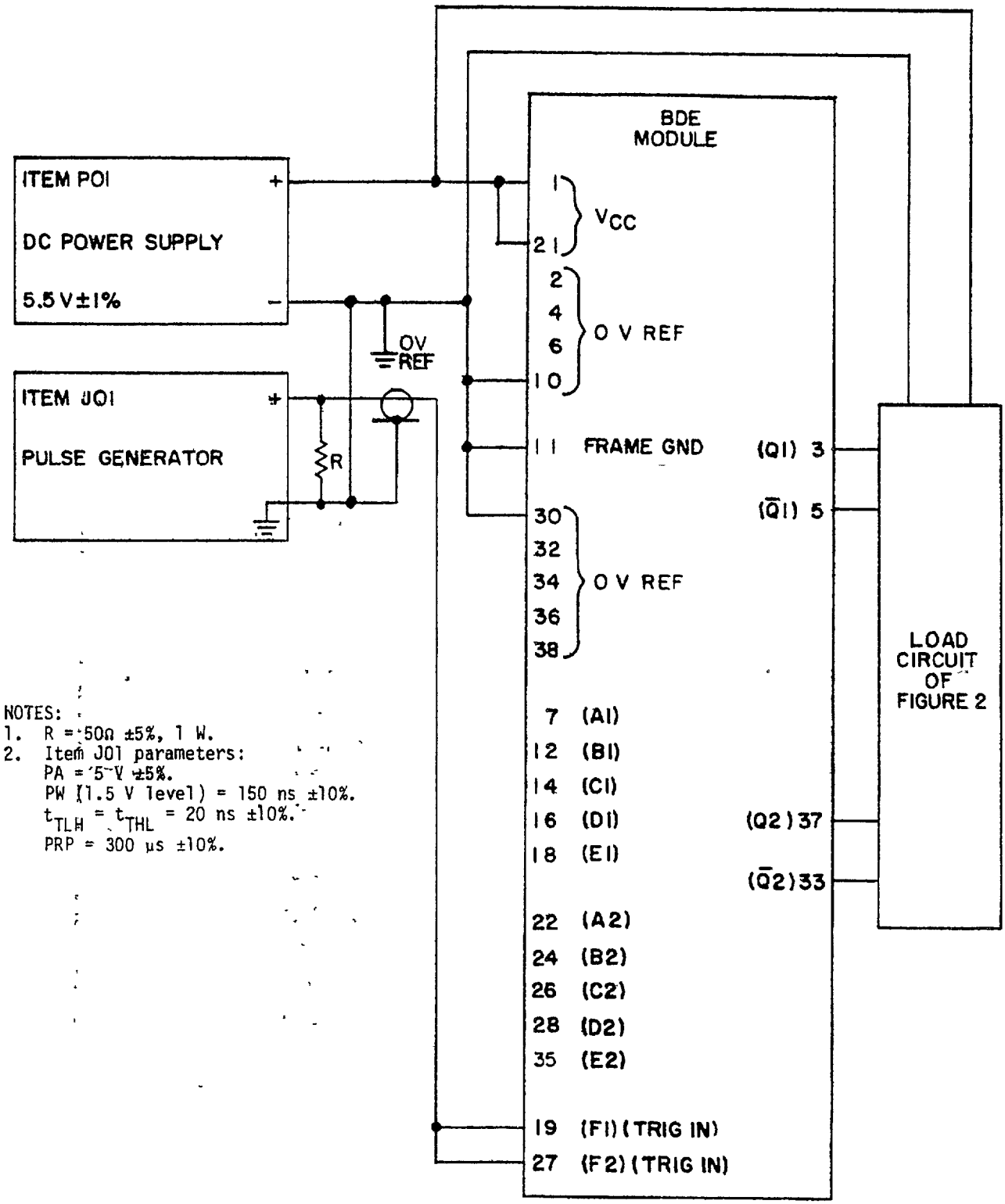
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NOTES:
1. R1 = R7 = 7.5 kΩ ±1%, 1/8 W.
2. R2 = R8 = 215Ω ±1%, 1/8 W.
3. R3 = R6 = R9 = R12 = 100Ω ±1%, 1/8 W.
4. R4 = R10 = 7.15 kΩ ±1%, 1/8 W.
5. R5 = R11 = 140Ω ±1%, 1/8 W.
6. CR1-CR16 = 1N4150 or equivalent.
7. C1 = C3 = C5 = C7 = 33 pF ±10%.
8. C2 = C4 = C6 = C8 = 330 pF ±10%.
9. C9 = 1.0 µF ±10%.
10. Loads shall be assembled on a board less than 3 by 3 (76.2 mm by 76.2 mm) with components and wiring within .50 inch (12.7 mm) of the board.
11. The maximum lead length from load to the appropriate module pin shall be 2.0 inches (50.8 mm).

FIGURE 2. BDE module load circuit.

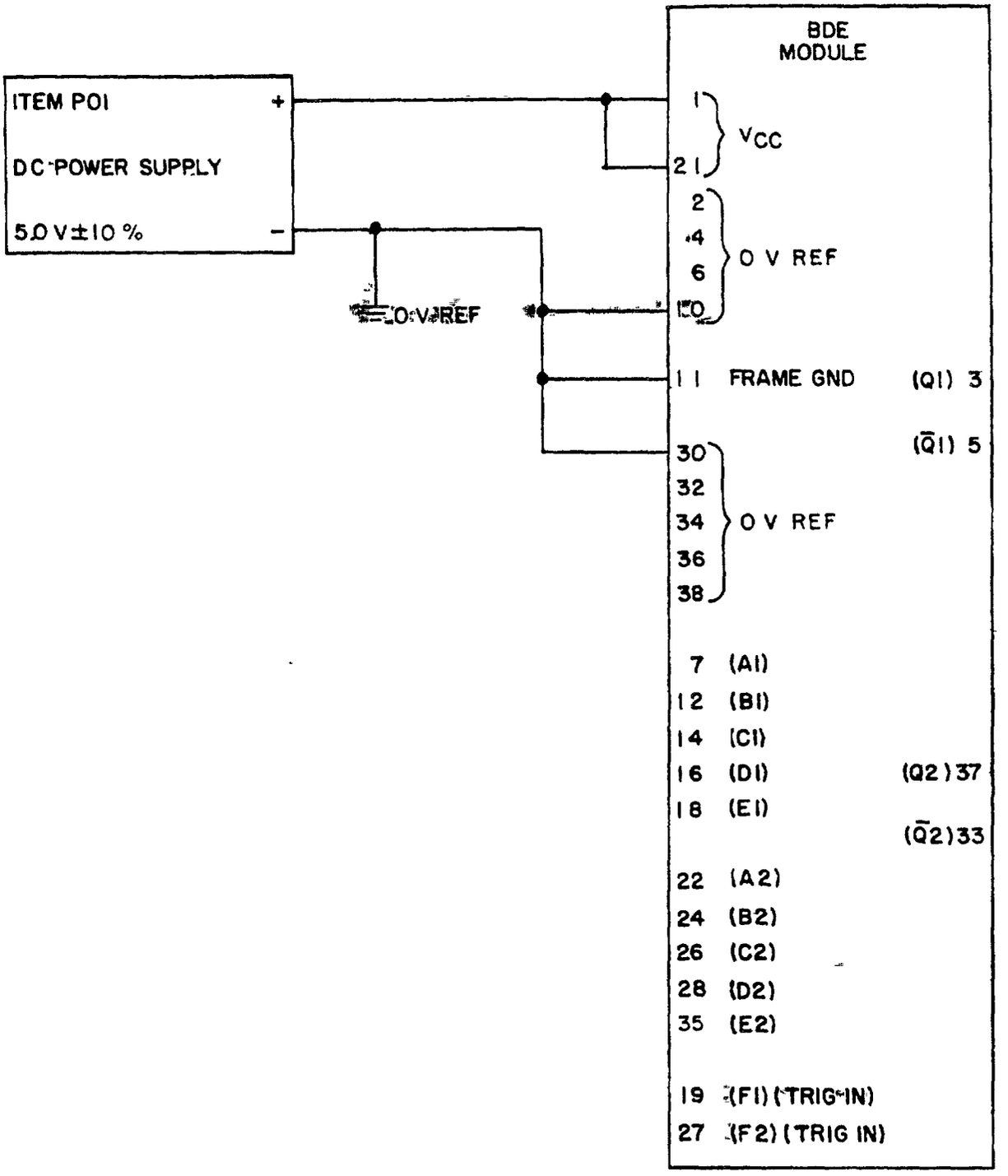
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NOTES:
1. R = 50Ω ±5%, 1 W.
2. Item J01 parameters:
   PA = 5 V ±5%.
   PW (1.5 V level) = 150 ns ±10%.
   $t_{TLH} = t_{THL}$ = 20 ns ±10%.
   PRP = 300 µs ±10%.

* FIGURE 3. Operating temperature, component temperature, and life test circuit.

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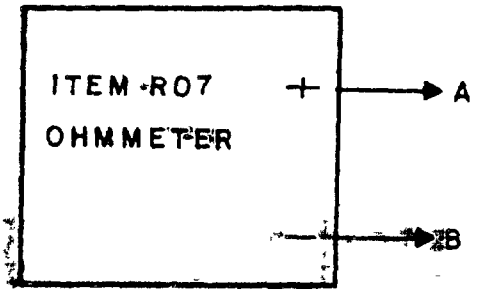
* FIGURE 4. Vibration test circuit.

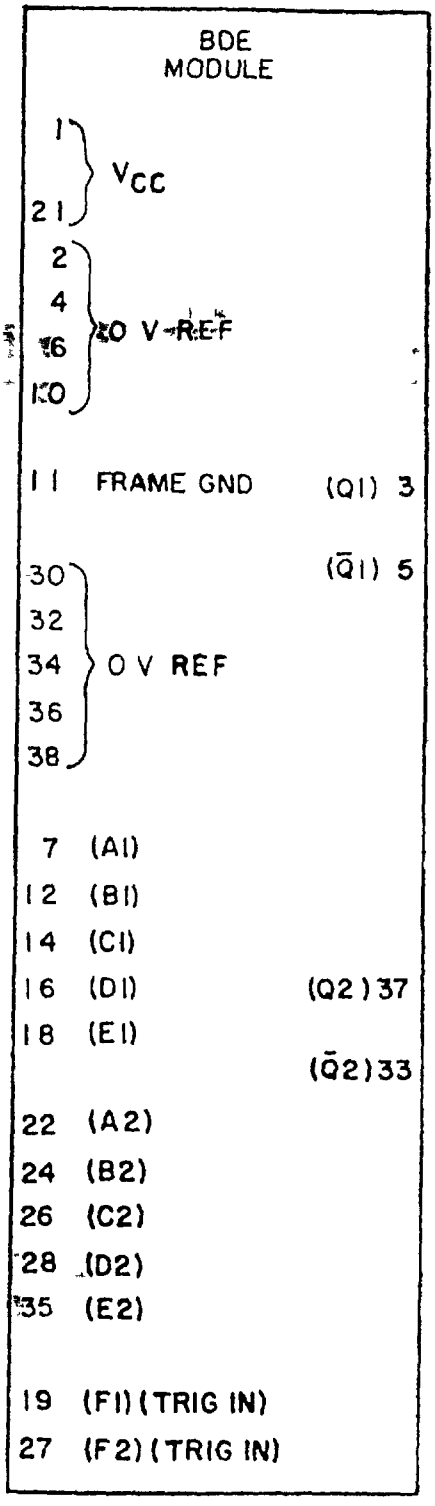
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* TABLE II. Continuity and isolation test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
A1	Continuity	10	2	0	1	Ω	5
A2	↓	10	4	↓	↓	↓	↓
A3		10	6				
A4		30	32				
A5		30	34				
A6		30	36				
A7		30	38				
A8		11	Frame				
B1	Isolation	11	All other pins interconnected	10	∞	MΩ	
B2	Isolation	Unused pins 8, 9,13,15,17,20, 23,25,29,31,39, and 40 interconnected	To all other pins interconnected	10	∞	MΩ	

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NOTE: For test condition A8 penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal.

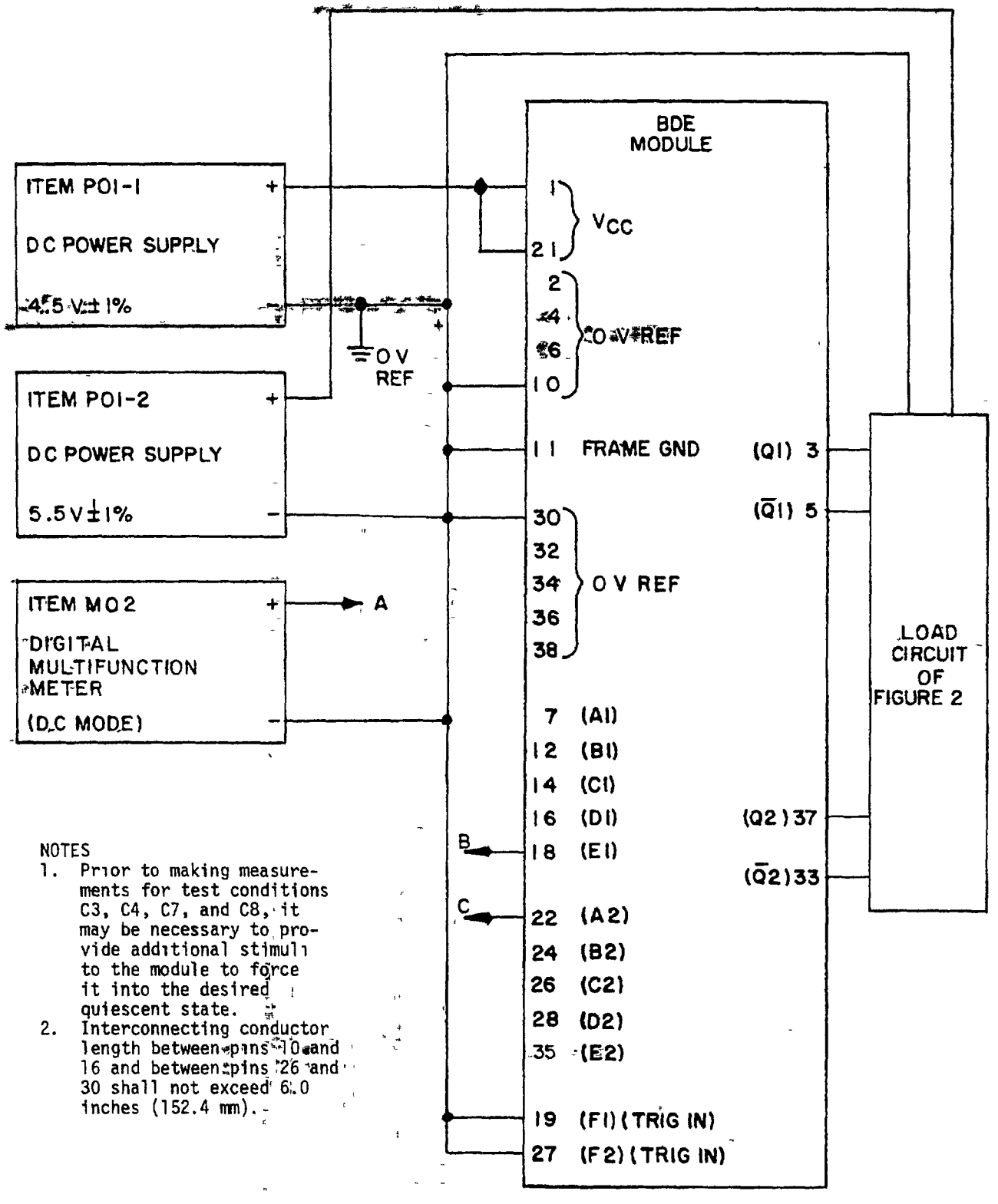
* FIGURE 5. Continuity and isolation test circuit for table II.

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* TABLE III. Output (low-level and high-level) test.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
C1	Q1 Output, low level	3	Open	Open	0.0	0.4	V	6
C2	$\overline{Q1}$ Output, high level	5	Open	↓	3.8	5.5	↓	↓
C3	Q1 Output, high level	3	10	↓	3.8	5.5	↓	↓
C4	$\overline{Q1}$ Output, low level	5	10	↓	0.0	0.4	↓	↓
C5	Q2 Output, low level	37	Open	↓	0.0	0.4	↓	↓
C6	$\overline{Q2}$ Output, high level	33	↓	↓	3.8	5.5	↓	↓
C7	Q2 Output, high level	37	↓	30	3.8	5.5	↓	↓
C8	$\overline{Q2}$ Output, low level	33	↓	30	0.0	0.4	↓	↓

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NOTES
1. Prior to making measurements for test conditions C3, C4, C7, and C8, it may be necessary to provide additional stimuli to the module to force it into the desired quiescent state.
2. Interconnecting conductor length between pins 10 and 16 and between pins 26 and 30 shall not exceed 6.0 inches (152.4 mm).

* FIGURE 6. Output (low-level and high-level) test circuit for table III.

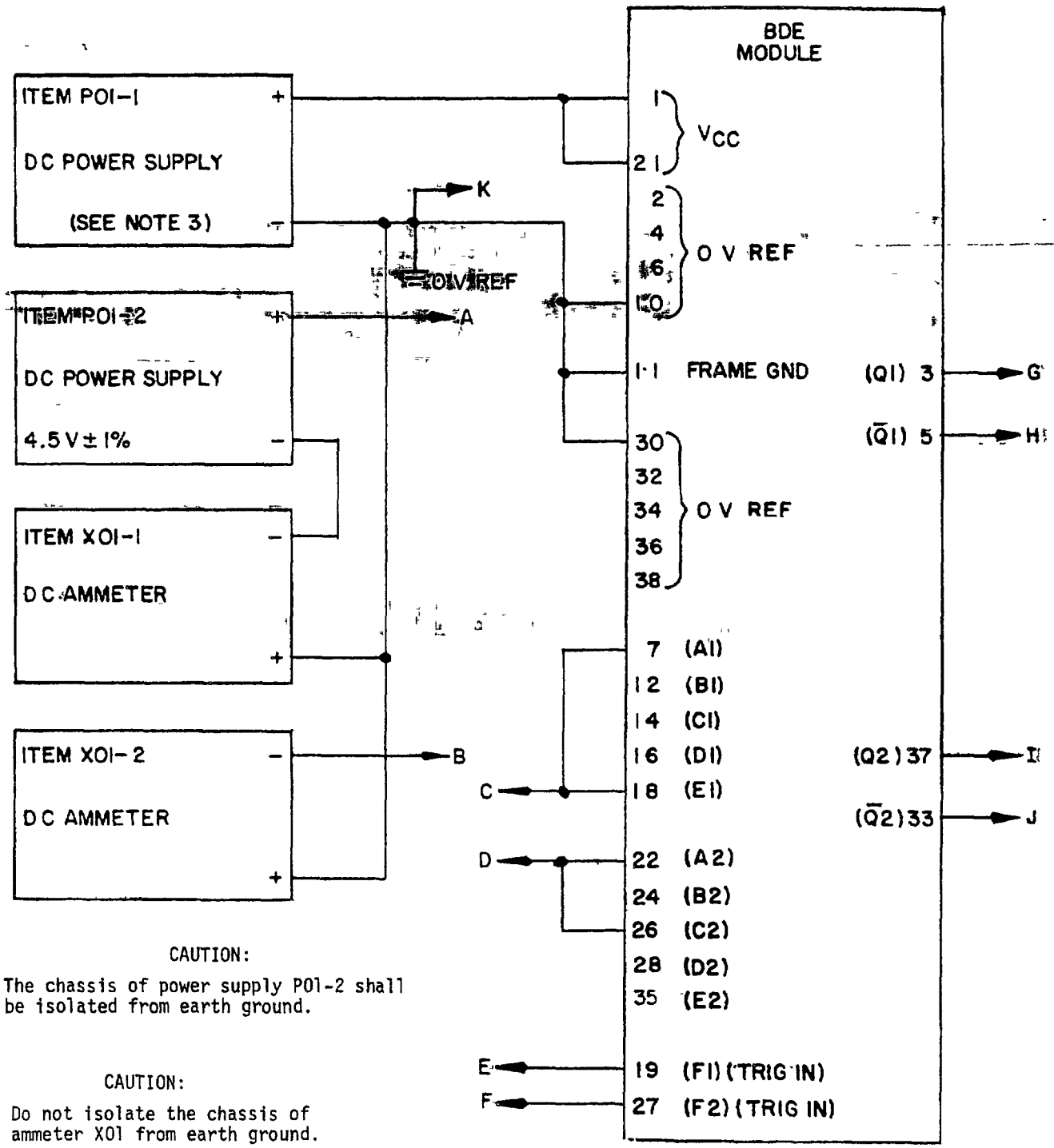
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* TABLE IV. Output short circuit current and input current (high-level and low-level) test.

Test condition	Parameter	Connect								Initial limits 25°±5°C ambient		Unit	Test figure
		C 1/ to pin	D 1/ to pin	E to pin	F to pin	G to pin	H to pin	I to pin	J to pin	Min	Max		
D1	Q1 Output short circuit current	12 and 10	24	K	K	B	Open	Open	Open	16	65	mA	7
D2	$\bar{Q}1$	12	24			Open	B	Open	Open				
D3	Q2		24 and 30				Open	B	Open				
D4	$\bar{Q}2$		24					Open	B				
D5	B1 and B2 Input current, high level			A	Open				Open	0.0	4.0	µA	
D6	B1 and B2 Input current, low level			B	Open						3.2	mA	
D7	B1 and B2 Input current, high level			Open	A						4.0	µA	
D8	B1 and B2 Input current, low level			Open	B						3.2	mA	

1/ Interconnection length between any two pins specified in these columns shall not exceed six inches (152.4 mm).

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* FIGURE 7. Output short circuit current and input current (high-level and low-level) test circuit for table IV.

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NOTES:
1. Prior to making measurements for test conditions D1 and D3, it may be necessary to provide additional stimuli to the module to force it into the desired quiescent state.
2. The conductor lengths between any two pins listed in the connect column of table IV shall not exceed 6.0 inches (152.4 mm).
3. Adjust item P01-1 voltage as follows:
   V_{CC} = 5.5 V ±1% for test conditions D1-D4, D6 and D8.
   V_{CC} = 4.5 V ±1% for test conditions D5 and D7.

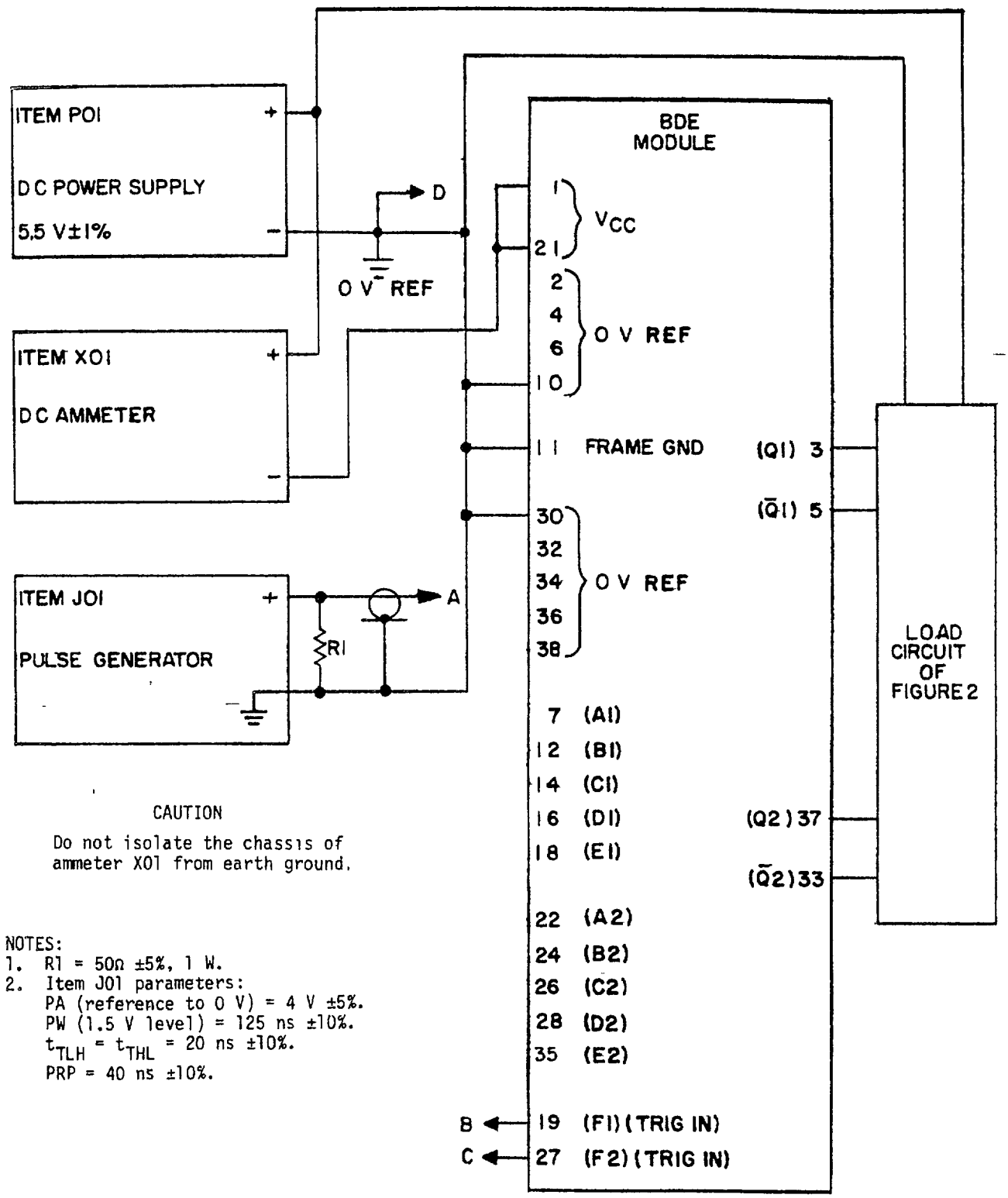
* FIGURE 7. Output short circuit current and input current (high-level and low-level) test circuit for table IV - Continued.

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* TABLE V. Power supply current test.

Test condition	Parameter	Connect		Initial limits 25° ±5°C ambient		Unit	Test figure
		B to pin	C to pin	Min	Max		
E1	Power supply current (quiescent)	D	D	0	120	mA	8
E2	Power supply current (operating)	A	A	0	120	mA	8

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CAUTION

Do not isolate the chassis of ammeter X01 from earth ground.

NOTES:
1. R1 = 50Ω ±5%, 1 W.
2. Item J01 parameters:
   PA (reference to 0 V) = 4 V ±5%.
   PW (1.5 V level) = 125 ns ±10%.
   $t_{TLH} = t_{THL}$ = 20 ns ±10%.
   PRP = 40 ns ±10%.

* FIGURE 8. Power supply current test circuit for table V.

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* TABLE VI. Output pulse width test.

Test condition	Parameter		Connect						Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	Min	Max		
F1	Q1	Output pulse width (10 µs)	12	16	--	--	19	3	9.3	10.7	µs	9,10
F2	$\overline{Q1}$		12	16	--	--	19	5				
F3	Q2		24	26	--	--	27	37				
F4	$\overline{Q2}$		24	26	--	--	27	33				
F5	Q1	Output pulse width (20 µs)	7	12	14	16	19	3	18.6	21.4		
F6	$\overline{Q1}$		7	12	14	16	19	5				
F7	Q2		24	35	26	28	27	37				
F8	$\overline{Q2}$		24	35	26	28	27	33				
F9	Q1	Output pulse width (30 µs)	14	16	--	--	19	3	27.9	32.1		
F10	$\overline{Q1}$		14	16	--	--	19	5				
F11	Q2		26	28	--	--	27	37				
F12	$\overline{Q2}$		26	28	--	--	27	33				
F13	Q1	Output pulse width (40 µs)	7	14	16	18	19	3	37.2	42.8		
F14	$\overline{Q1}$		7	14	16	18	19	5				
F15	Q2		28	35	22	26	27	37				
F16	$\overline{Q2}$		28	35	22	26	27	33				
F17	Q1	Output pulse width (50 µs)	12	14	16	18	19	3	46.5	53.5		
F18	$\overline{Q1}$	Output pulse width (50 µs)	12	14	16	18	19	5	46.5	53.5		

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* TABLE VI. Output pulse width test - Continued.

Test condition	Parameter		Connect						Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	Min	Max		
F19	Q2	Output pulse width (50 µs)	24	28	22	26	27	37	46.5	53.5	µs	9,10
F20	$\overline{Q}$2	Output pulse width (50 µs)	24	28	22	26	27	33	46.5	53.5		
F21	Q1	Output pulse width (60 µs)	7	12	16	18	19	3	55.8	64.2		
F22	$\overline{Q}$1	↓	7	12	16	18	19	5	↓	↓		
F23	Q2		24	35	22	26	27	37				
F24	$\overline{Q}$2		24	35	22	26	27	33				
F25	Q1	Output pulse width (70 µs)	16	18	--	--	19	3	65.1	74.9		
F26	$\overline{Q}$1	↓	16	18	--	--	19	5	↓	↓		
F27	Q2		22	26	--	--	27	37				
F28	$\overline{Q}$2		22	26	--	--	27	33				
'29	Q1	Output pulse width (80 µs)	7	18	--	--	19	3	74.4	85.6		
F30	$\overline{Q}$1	↓	7	18	--	--	19	5	↓	↓		
F31	Q2		22	35	--	--	27	37				
F32	$\overline{Q}$2		22	35	--	--	27	33				
F33	Q1	Output pulse width (90 µs)	12	18	--	--	19	3	83.7	96.3		
F34	$\overline{Q}$1	↓	12	18	--	--	19	5	↓	↓		
F35	Q2		22	24	--	--	27	37				
F36	$\overline{Q}$2		22	24	--	--	27	33				

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* TABLE VI. Output pulse width test - Continued.

Test condition	Parameter	Connect						Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	Min	Max		
F37	Q1 Output pulse width (100 µs)	14	18	7	12	19	3	93	107	µs	9,10
F38	$\overline{Q1}$	14	18	7	12	19	5				
F39	Q2	24	35	22	28	27	37				
F40	$\overline{Q2}$	24	35	22	28	27	33				
F41	Q1 Output pulse width (110 µs)	14	18	--	--	19	3	102	118		
F42	$\overline{Q1}$	14	18	--	--	19	5				
F43	Q2	22	28	--	--	27	37				
F44	$\overline{Q2}$	22	28	--	--	27	33				
F45	Q1 Output pulse width (120 µs)	7	14	--	--	19	3	112	128		
F46	$\overline{Q1}$	7	14	--	--	19	5				
F47	Q2	28	35	--	--	27	37				
F48	$\overline{Q2}$	28	35	--	--	27	33				
F49	Q1 Output pulse width (130 µs)	12	14	--	--	19	3	121	139		
F50	$\overline{Q1}$	12	14	--	--	19	5				
F51	Q2	24	28	--	--	27	37				
F52	$\overline{Q2}$	24	28	--	--	27	33				
F53	Q1 Output pulse width (140 µs)	7	12	--	--	19	3	130	150		
F54	$\overline{Q1}$ Output pulse width (140 µs)	7	12	--	--	19	5	130	150		

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* TABLE VI. Output pulse width test - Continued.

Test condition	Parameter	Connect						Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	Min	Max		
F55	Q2 Output pulse width (140 µs)	24	35	--	--	27	37	130	150	µs	9,10
F56	$\overline{Q}$2 Output pulse width (140 µs)	24	35	--	--	27	33	130	150		
F57	Q1 Output pulse width (150 µs)	--	--	--	--	19	3	139	161		
F58	$\overline{Q}$1	--	--	--	--	19	5				
F59	Q2	--	--	--	--	27	37				
F60	$\overline{Q}$2	--	--	--	--	27	37				

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* TABLE VII. Transition time (t_{THL} and t_{TLH}) test.

Test condition	Parameter	Connect 1/		Initial limits 25°±5°C ambient		Unit	Test figure
		E to pin	F to pin	Min	Max		
G1	Q1 Transition time (t_{THL})	19	3	3	40	ns	9,10
G2	$\overline{Q}$1	19	5				
G3	Q2	27	37				
G4	$\overline{Q}$2	27	33				
G5	$\overline{Q}$2	27	33				
G6	Q1 Transition time (t_{TLH})	19	3	10	250		
G7	Q1	19	3				
G8	$\overline{Q}$1	19	5				
G9	Q2	27	37				
G10	$\overline{Q}$2	27	33				

1/ Connect pin 12 to pin 16 and pin 24 to pin 26 for all test conditions, except test conditions G5 and G7.

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* TABLE VIII. Propagation delay time (t_{PLH} and t_{PHL}) test.

Test condition	Parameter	Connect 1/		Initial limits 25°±5°C ambient		Unit	Test figure
		E to pin	F to pin	Min	Max		
H1	Q1 Propagation delay time (t_{PLH})	19	3	10	170	ns	9,10
H2	Q1 Propagation delay time (t_{PLH})	19	3	10			
H3	$\overline{Q}$1 Propagation delay time (t_{PHL})	19	5	7			
H4	Q2 Propagation delay time (t_{PLH})	27	37	10			
H5	$\overline{Q}$2 Propagation delay time (t_{PHL})	27	33	7			
H6	$\overline{Q}$2 Propagation delay time (t_{PHL})	27	33	7			

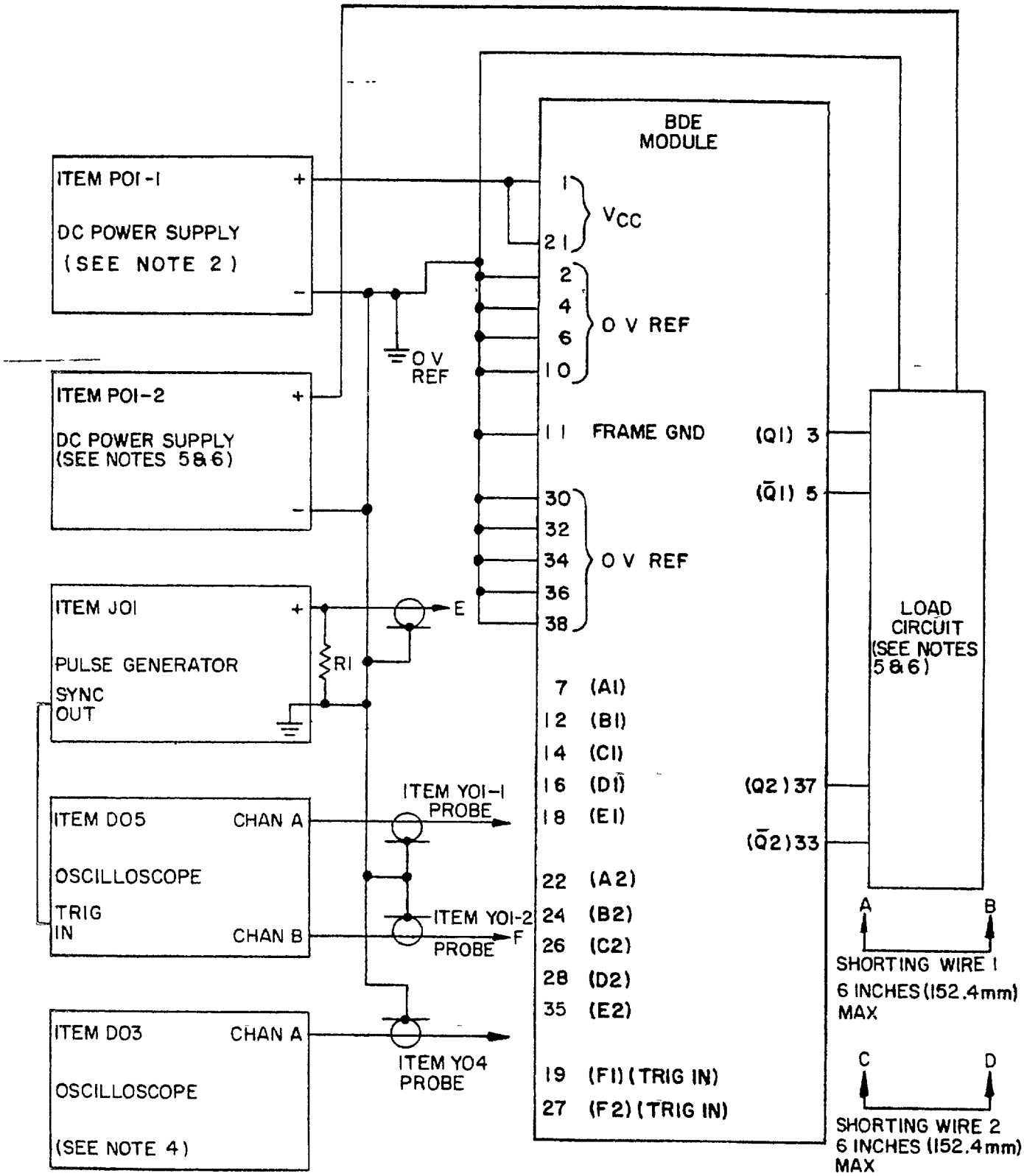
1/ Connect pin 12 to pin 16 and pin 24 to pin 26 for all test conditions, except test conditions H2 and H6.

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* TABLE IX. Output pulse amplitude (low-level and high-level), overshoot, undershoot and preshoot test.

Test condition	Parameter		Connect		Initial limits 25°±5°C ambient		Unit	Test figure
			E to pin	F to pin	Min	Max		
J1	Q1	Output pulse amplitude, low level	19	3	0.0	0.4	V	9,10
J2	$\overline{Q1}$		19	5				
J3	Q2		27	37				
J4	$\overline{Q2}$		27	33				
J5	Q1	Output pulse amplitude, high level	19	3	3.8	5.5		
J6	$\overline{Q1}$		19	5				
J7	Q2		27	37				
J8	$\overline{Q2}$		27	33				
J9	Q1	Overshoot	19	3	0.0	0.5		
J10	$\overline{Q1}$		19	5				
J11	Q2		27	37				
J12	$\overline{Q2}$		27	33				
J13	Q1	Undershoot	19	3				
J14	$\overline{Q1}$		19	5				
J15	Q2		27	37				
J16	$\overline{Q2}$		27	33				
J17	Q1	Preshoot	19	3				
J18	$\overline{Q1}$		19	5				
J19	Q2		27	37				
J20	$\overline{Q2}$		27	33				

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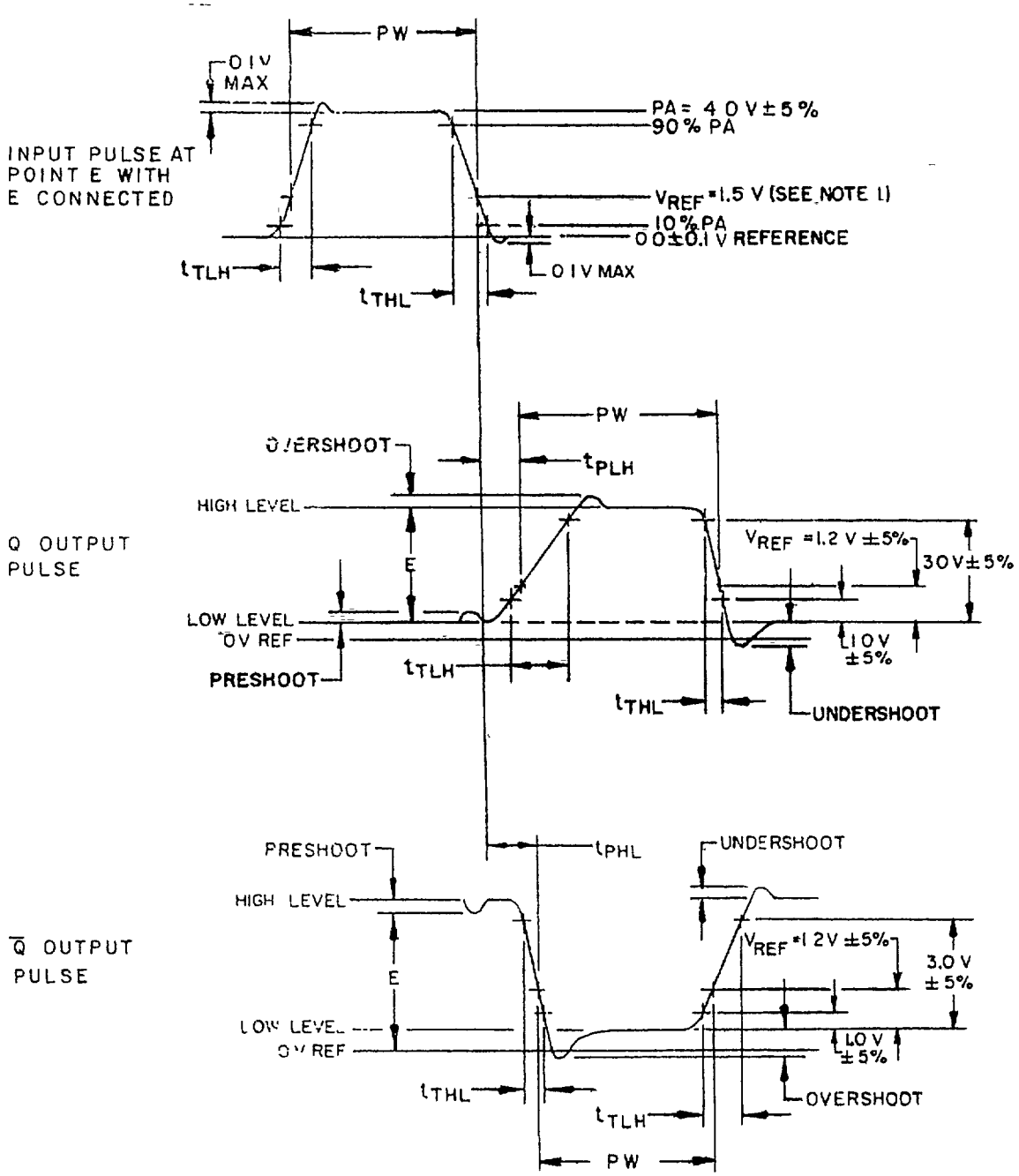
* FIGURE 9. Output pulse width, transition time (t_{THL} and t_{TLH}), propagation delay time (t_{PLH} and t_{PHL}), output pulse amplitude (low-level and high-level), overshoot, undershoot, and preshoot test circuit for tables VI through IX.

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NOTES:
1. R1 = 50Ω ±5%, 1 W.
2. Set item P01-1 voltage as follows:
   V_{CC} = 5.5 V ±1% for test conditions F1, F4, F5, F8, F10, F11, F13, F16, F18, F19, F22, F23, F26, F27, F29, F32, F34, F35, F38, F39, F42, F43, F46, F47, F50, F51, F54, F55, F57 and F60.
   = 4.5 V ±1% for test conditions F2, F3, F6, F7, F9, F12, F14, F15, F17, F20, F21, F24, F25, F28, F30, F31, F33, F36, F37, F40, F41, F44, F45, F48, F49, F52, F53, F56, F58, F59, G1-G10, H1-H6, and J1-J20.
3. For test conditions G1-G10, H1-H6 and J1-J20, connect probe Y01-2 as close as physically possible to the module pin under test.
4. For test conditions J1-J8, use item D03 with item Y04; measurements must have 5 mV/mm resolution.
5. Use load circuit of figure 2 for test conditions F1-F60 and set item P01-2 to 5.5 V ±1%.
6. Use load circuit of figure 11 for test conditions G1-G10, H1-H6 and J1-J20 and set item P01-2 to 2.5 V ±1%.
7. All connections from load circuit to module output pins shall be 2 inches (50.8 mm) maximum.

* FIGURE 9. Output pulse width, transition time (t_{THL} and t_{TLH}), propagation delay time (t_{PLH} and t_{PHL}), output pulse amplitude (low-level and high-level), overshoot, undershoot, and preshoot test circuit for tables VI through IX - Continued.

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* FIGURE 10. Output pulse width, transition time (t_{THL} and t_{TLH}), propagation delay time (t_{PLH} and t_{PHL}), output pulse amplitude (low-level and high-level), overshoot, undershoot, preshoot, input sensitivity and immunity test circuit waveforms for tables VI through IX and XI.

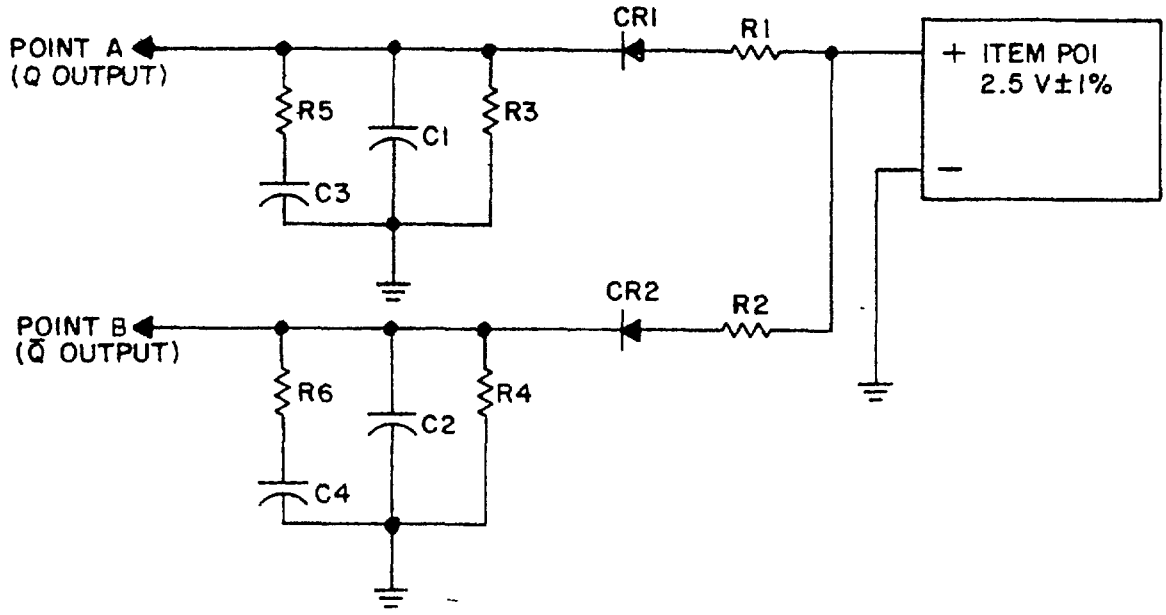
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NOTES:

1. Several perturbations which may occur on the output waveform are shown (preshoot, overshoot, undershoot). Perturbations may occur at other locations on the output waveforms, but they must conform to the following requirements. No negative perturbations on the high level or positive perturbations on the low level shall exceed 0.5 V amplitude.
2. Input pulse parameters:
   PW (50% PA) = 125 ns ±10% for test conditions J1-J12.
   PW (1.5 V level) = 125 ns ±10% for test conditions F1-F8, F13-F16, F29-F32, F57-F60, G1-G10, H1-H6 and J13-J20.
   PW (1.5 V level) = 100 µs ±10% for test conditions F9-F12, F17-F28, and F33-F56.
   $t_{TLH} = t_{THL}$ = 25 ns ±5%.
   PRP = 300 µs ±10%.

* FIGURE 10. Output pulse width, transition time (t_{THL} and t_{TLH}), propagation delay time (t_{PLH} and t_{PHL}), output pulse amplitude (low-level and high-level), overshoot, undershoot, preshoot, input sensitivity and immunity test circuit waveforms for tables VI through IX and XI - Continued.

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NOTES:
1. R1 = Select value for a current flow of 35.2 mA ±1% into point A when point A is held at 0.4 V.
2. R2 = Select value for a current flow of 22.4 mA ±1% into point B when point B is held at 0.4 V.
3. This load shall be applied to each module circuit Q and $\overline{Q}$ output.
4. R3 = R4 = 18.7 kΩ ±1%, 1/8 W.
5. R5 = R6 = 100Ω ±1%, 1/8 W.
6. C1 = C2 = 33 pF ±10%.
7. C3 = C4 = 330 pF ±10%.
8. CR1 = CR2 = 1N4150 or equivalent.

* FIGURE 11. BDE logic load circuit.

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* TABLE X. Pulse width (recovery time) test.

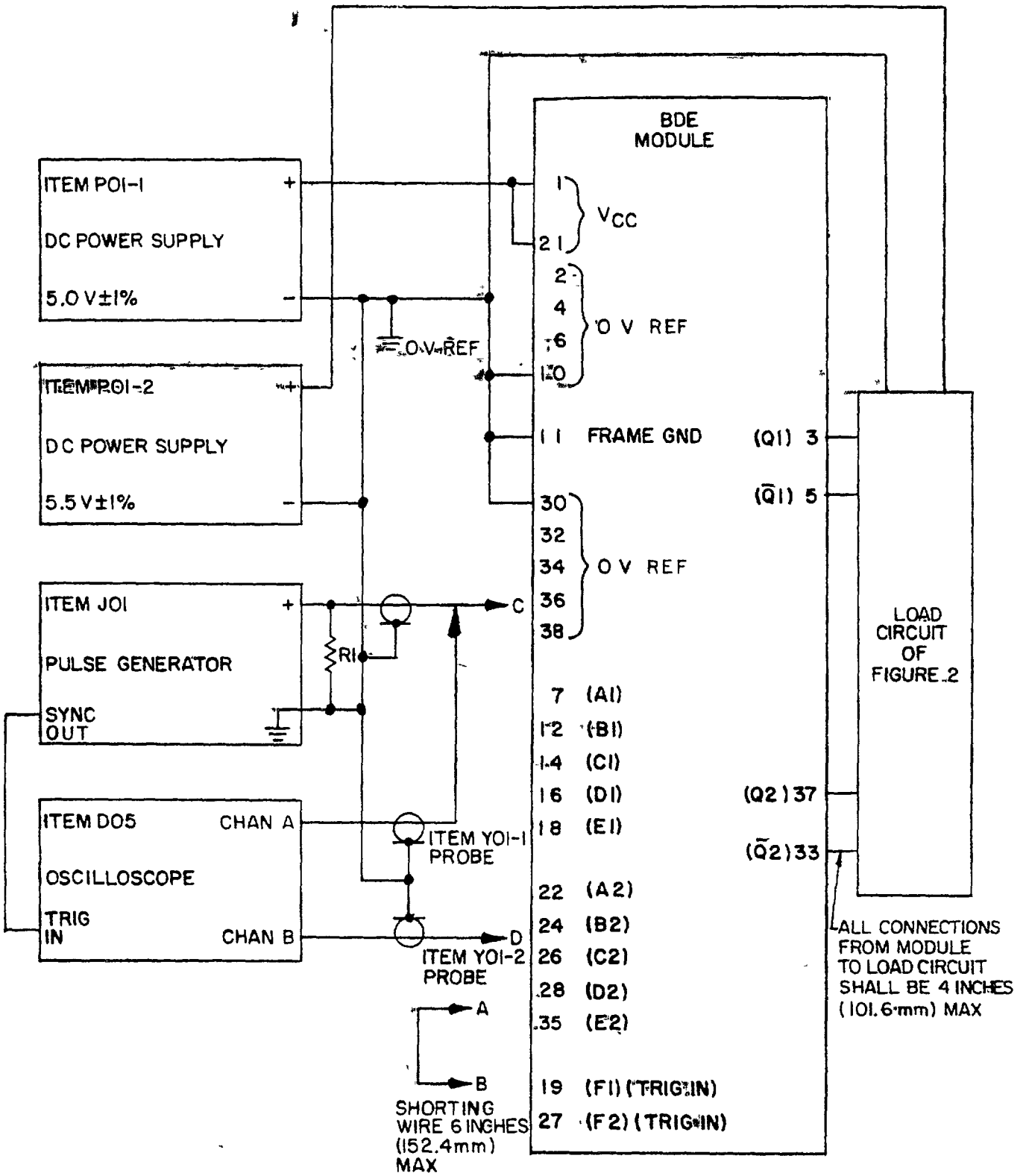
Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
K1	Q1 Pulse width (recovery time)	12	16	19	3	9.3	10.7	µs	12
K2	Q1	Open	Open	19	3	139	161		
K3	Q2	24	26	27	37	9.3	10.7		
K4	Q2	Open	Open	27	37	139	161		

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* TABLE XI. Input sensitivity and immunity test.

Test condition	Parameter		Connect				Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
L1	Q1	Input trigger sensitivity	12	16	19	3	9.3	10 7	µs	10,12
L2	$\overline{Q1}$		12	16	19	5				
L3	Q2		24	26	27	37				
L4	$\overline{Q2}$		24	26	27	33				
L5	Q1	Input immunity (20 ns)	12	16	19	3	0	5	ns	
L6	$\overline{Q1}$		12	16	19	5				
L7	Q2		24	26	27	37				
L8	$\overline{Q2}$		24	26	27	33				
L9	Q1	Input immunity (0 95 V)	12	16	19	3				
L10	$\overline{Q1}$		12	16	19	5				
L11	Q2		24	26	27	37				
L12	$\overline{Q2}$		24	26	27	33				

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* FIGURE 12. Pulse width (recovery time), input sensitivity, and immunity test circuit for tables X and XI.

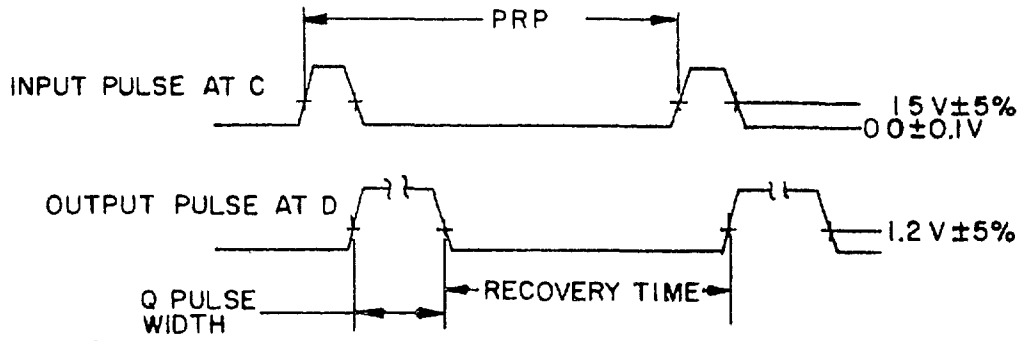
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NOTES:
1. R1 = 50Ω ±5%, 1 W.
2. Input pulse parameters (test conditions K1 and K3):

   a. PA (reference to 0 V) = 4 V ±5%.
   b. Pulse width (1.5 V level) = 125 ns ±10%.
   c. $t_{TLH} = t_{THL}$ = 20 ns ±10%.
   d. PRP = 36 µs ±5%.

3 Input pulse parameters (test conditions K2 and K4):

   a. PA (reference to 0 V) = 4 V ±5%.
   b. Pulse width (1.5 V level) = 125 ns ±10%.
   c. $t_{TLH} = t_{THL}$ = 20 ns ±10%.
   d. PRP = 176 µs ±5%.



4. Input pulse parameters (test conditions L1 through L4):

   a. PA (reference to 0 V) = 3.2 V ±5%.
   b Pulse width (1.5 V level) = 135 ns ±5%
   c. $t_{TLH} = t_{THL}$ = 40 ns ±5%
   d. PRP = 20 kHz ±10%.

5. Input pulse parameters (test conditions L5 through L8):

   a. PA (reference to 0 V) = 5.0 V ±5%.
   b. Pulse width (1.5 V level) = 20 ns ±5%.
   c. $t_{TLH} = t_{THL}$ = 8 ns ±5%.
   d. PRP = 20 kHz ±10%.

6. Input pulse parameters (test conditions L9 through L12):

   a. PA (reference to 0 V) = 0.95 V ±5%.
   b. Pulse width (1.5 V level) = 1000 ns ±5%.
   c. $t_{TLH} = t_{THL}$ = 8 ns ±5%.
   d. PRP = 20 kHz ±10%.

7. For test conditions L5 through L12, if a pulse is present at the output of the module (point D), it shall not exceed the limits of table XI

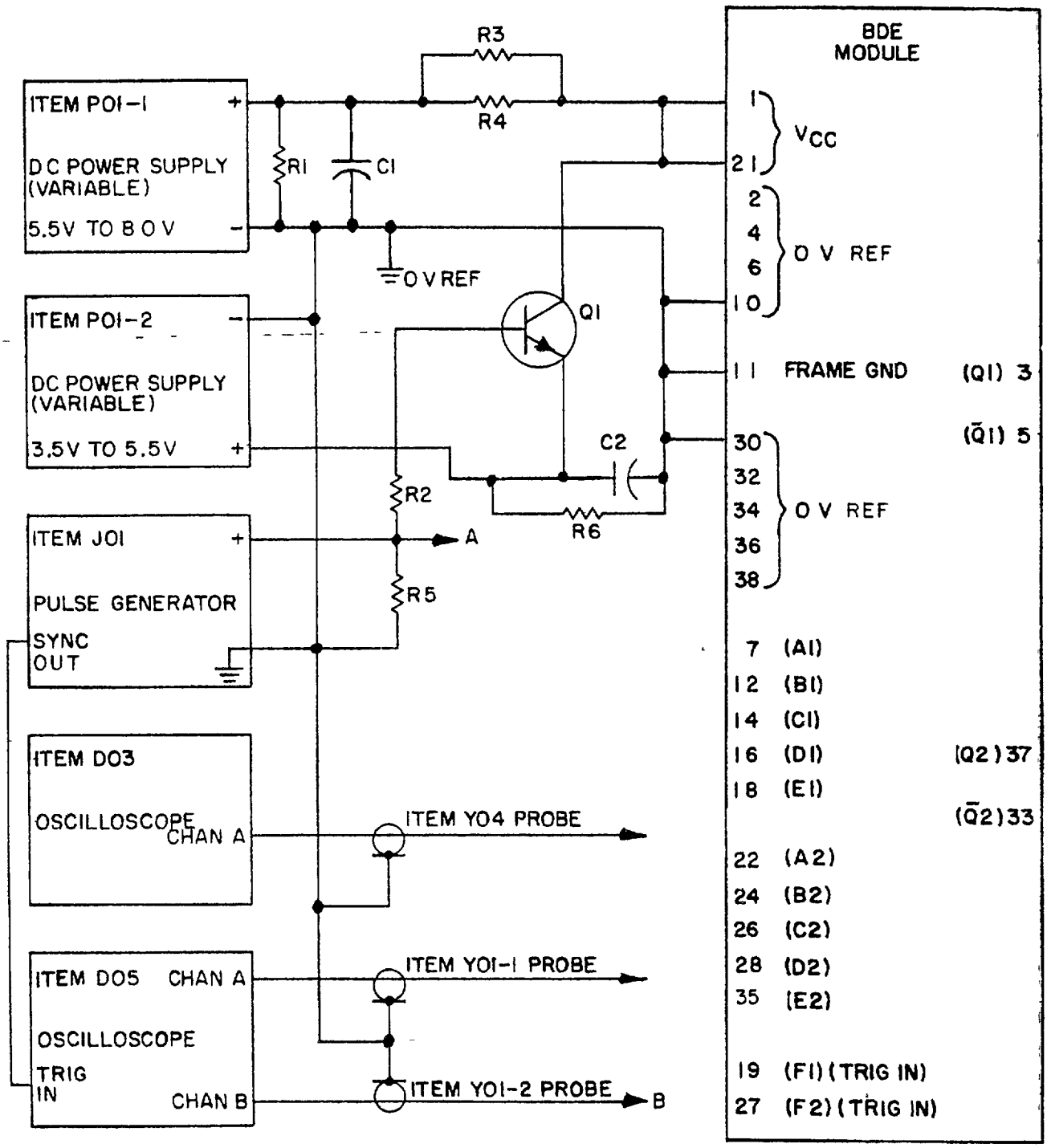
* FIGURE 12. Pulse width (recovery time), input sensitivity, and immunity test circuit for tables X and XI - Continued.

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* TABLE XII. Power supply (transient rise time and false trigger) test.

Test condition	Parameter		Connect B to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
M1	$\overline{Q}1$	Power supply, transient rise time	5	9	53	µs	13
M2	$\overline{Q}2$	↓	33	9	53	↓	↓
M3	$\overline{Q}1$	↓	5	1.5	40.0	↓	↓
M4	$\overline{Q}2$	↓	33	1.5	40.0	↓	↓
M5	$\overline{Q}1$	Power supply, false trigger	5	2.0	5.5	V	↓
M6	$\overline{Q}2$	↓	33	↓	↓	↓	↓
M7	$\overline{Q}1$	↓	5	↓	↓	↓	14
M8	$\overline{Q}2$	↓	33	↓	↓	↓	14

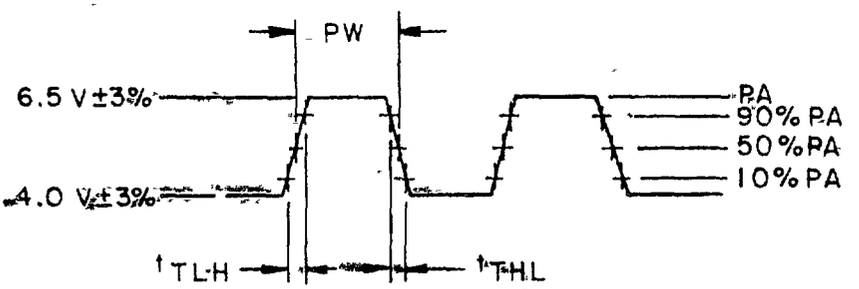
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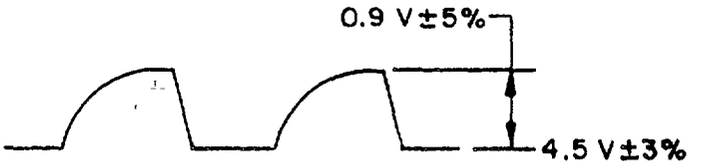
* FIGURE 13. Power supply (transient rise time and false trigger) test circuit for table XII.

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NOTES:
1. R1 = R6 = 18.2Ω ±1%, 3 W.
2. R2 = R5 = 51.1Ω ±1%, 1/2 W.
3. R3 = R4 = 10Ω ±1%, 1 W.
4. C1 = C2 = 1.0 µF ±10%.
5. Q1 = 2N3108A or equivalent.

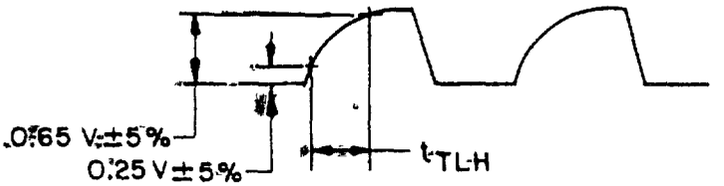


6. Using item D03, adjust item J01 to obtain the preceding waveform at point A with the output of item J01 disconnected from R2.
   a. $t_{TLH} = t_{THL}$ = 50 ns ±20%.
   b. PW (at 50% amplitude) = 400 µs ±10%.
   c. Duty cycle = 50%.



VOLTAGE PULSE AT MODULE PIN 1 (MODULE PLUGGED IN)

7. For test conditions M1 through M6, connect item J01 to R2; connect item Y04 to module pin 1; and adjust item P01-2 to set the baseline of the voltage pulse at module pin 1 (module plugged in) to 4.5 V ±3%. Then adjust item P01-1 to obtain a peak to peak amplitude of 0.9 V ±5%.



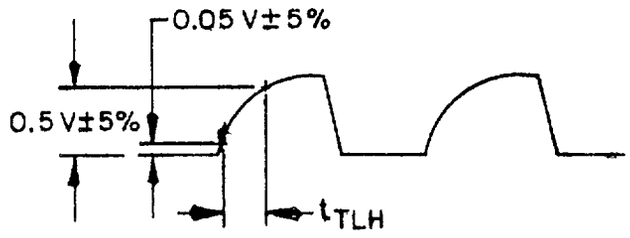
OUTPUT PULSE

8. For test conditions M1 and M2, the output pulse rise time shall be measured as shown above and shall conform to the requirements of table XII using item J05 and item Y01-2 (with 1X attenuator and AC coupler).

* FIGURE 13. Power supply (transient rise time and false trigger) test circuit for table XII - Continued.

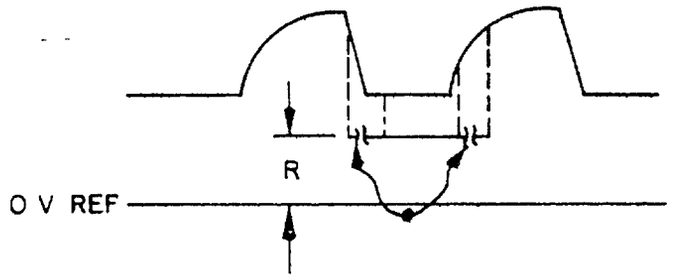
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9. For test conditions M3 and M4, the output pulse rise time shall be measured as shown below and shall conform to the requirements of table XII using item D05 and item Y01-2 (with 1X attenuator and AC coupler).



OUTPUT PULSE

10. For test conditions M5 and M6, the output pulse measurement (R) shall conform to the requirements of table XII using item D03. The value R is defined as the minimum voltage of the waveform.

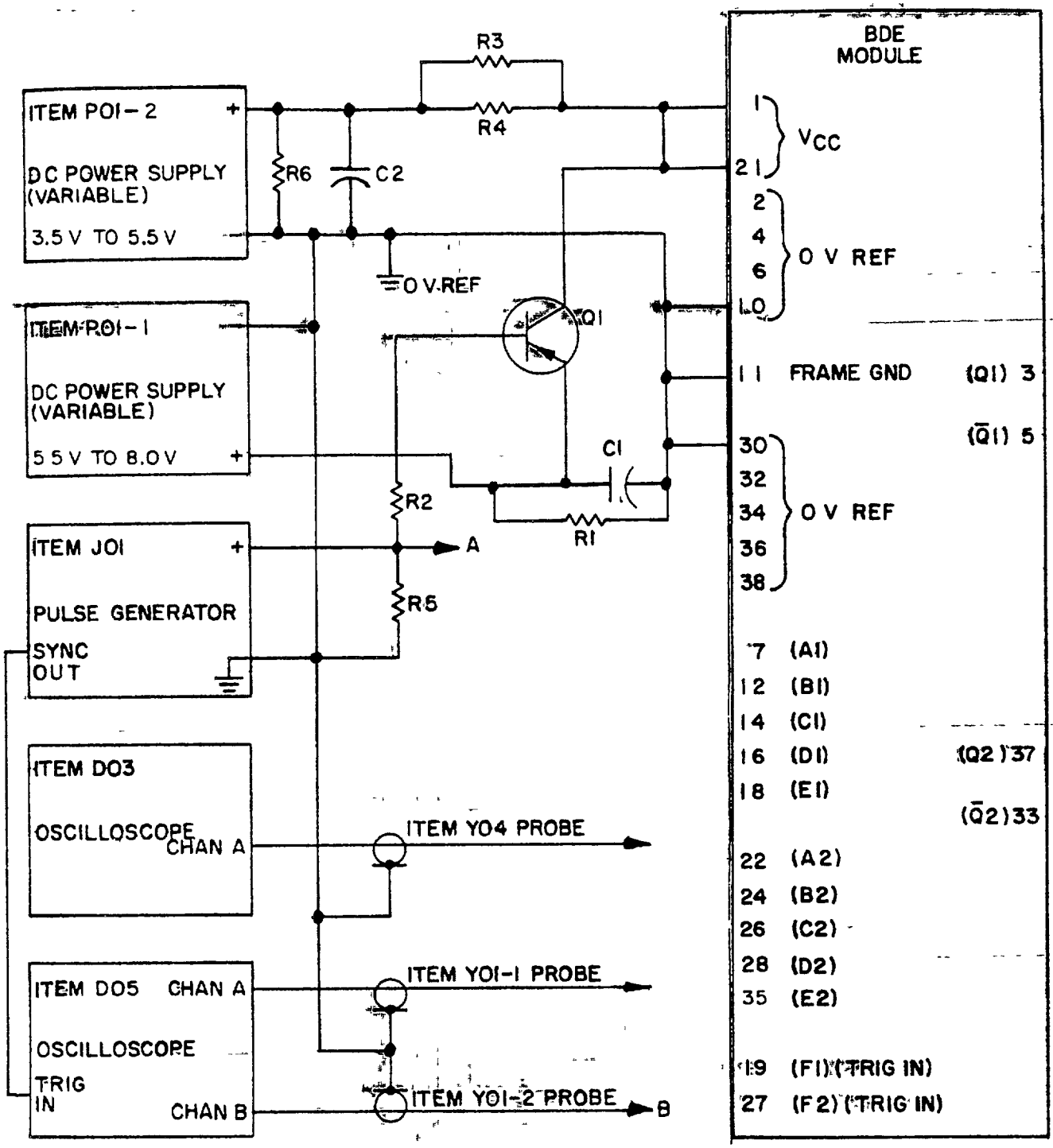


MODULE FALSE TRIGGERING MAY OCCUR AT EITHER OF THESE TWO POINTS

OUTPUT PULSE

* FIGURE 13. Power supply (transient rise time and false trigger) test circuit for table XII - Continued.

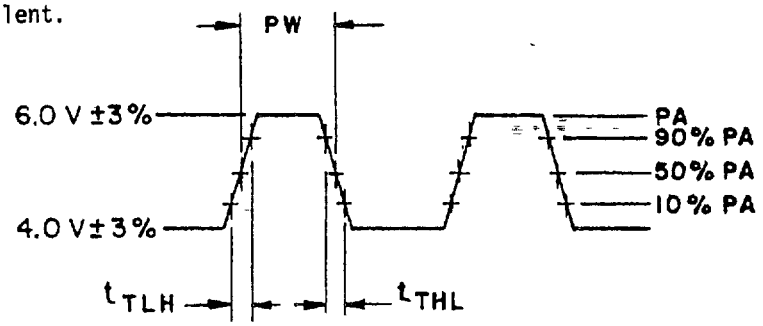
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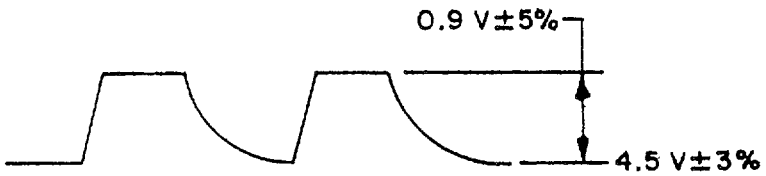
* FIGURE 14. Power supply (false trigger) test circuit for table XII.

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NOTES:
1. R1 = R6 = 18.2Ω ±1%, 3 W.
2. R2 = R5 = 51.1Ω ±1%, 1/2 W.
3. R3 = R4 = 10Ω ±1%, 1 W.
4. C1 = C2 = 1.0 µF ±10%.
5. Q1 = 2N4029 or equivalent.

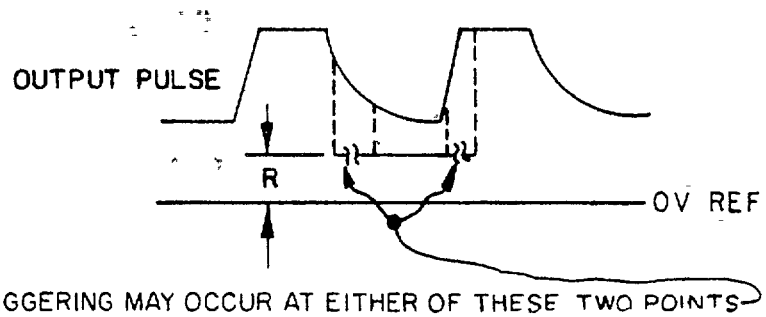


6. Using item D03, adjust item J01 to obtain the preceding waveform at point A with the output of item J01 disconnected from R2.
   a. $t_{TLH} = t_{THL}$ = 50 ns ±10%.
   b. PW (at 50% amplitude) = 400 µs ±10%.
   c. Duty cycle = 50%.



VOLTAGE PULSE AT MODULE PIN 1 (MODULE PLUGGED IN)

7. For test conditions M7 and M8, connect item J01 to R2; connect item Y04 to module pin 1; and adjust item P01-2 to set the baseline of the voltage pulse at module pin 1 (module plugged in) to 4.5 V ±3%. Then adjust item P01-1 to obtain a peak to peak amplitude of 0.9 V ±5%.



8. For test conditions M7 and M8, the output pulse measurement (R) shall conform to the requirements of table XII using item D03. The value R is defined as the minimum voltage of the waveform XII.

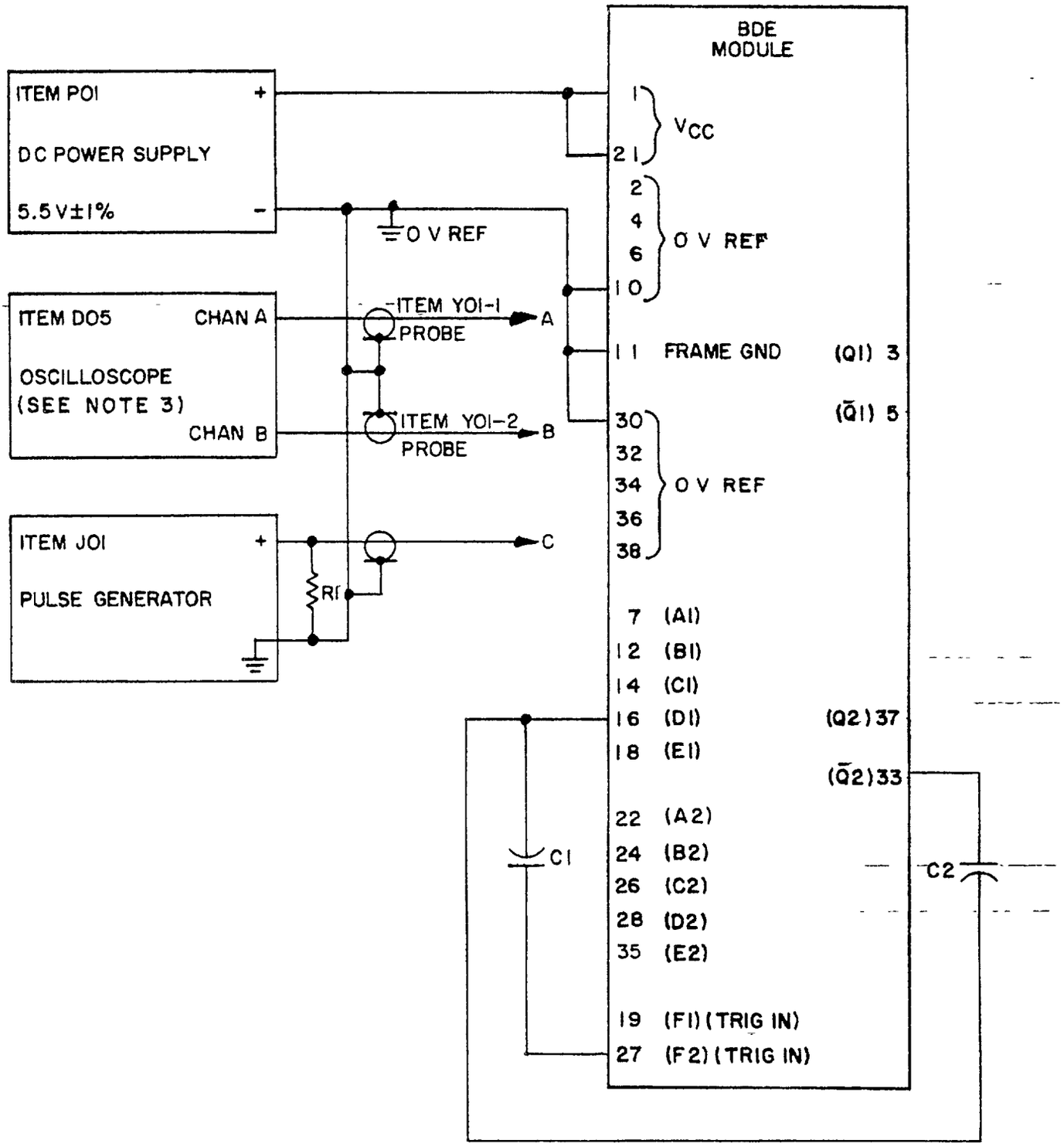
* FIGURE 14. Power supply (false trigger) test circuit for table XII - Continued.

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* TABLE XIII. Crosstalk test.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
N1	Q1 Crosstalk	27	3	27	0	5	ns	15,17
N2	Q1	33	3					
N3	Q1	33	3					
N4	$\overline{Q}$1	27	5					
N5	$\overline{Q}$1	33	5					
N6	$\overline{Q}$1	33	5					
N7	Q2	19	37	19				16,17
N8	Q2	5	37					
N9	Q2	5	37					
N10	$\overline{Q}$2	19	33					
N11	$\overline{Q}$2	5	33					
N12	$\overline{Q}$2	5	33					

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* FIGURE 15. Crosstalk test circuit for table XIII.

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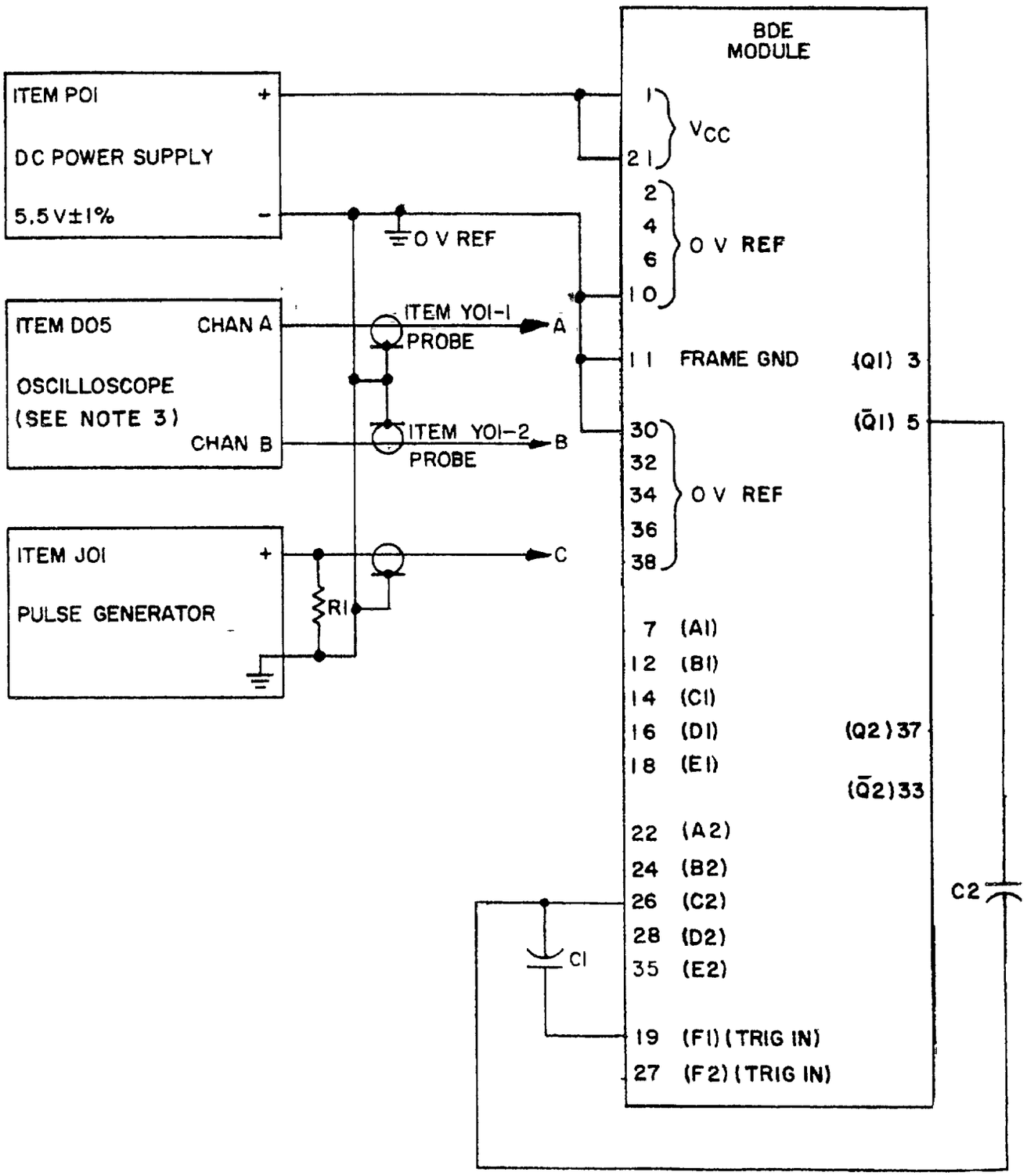
NOTES:
1. R1 = 50Ω ±5%, 1 W.
2. The following capacitances are measured using item CO2 at the test fixture pins with the module removed, at a frequency of 1 MHz ±10%, and will include stray wiring capacitance but not oscilloscope probe capacitance. Glass trimmer capacitors may be added to conform to the required values.

   a. C1 = 4 pF maximum.
   b. C2 = 5 ±1 pF.

3. Set the internal trigger slope of item DO5 to negative for test conditions N1, N3, N4 and N6 and positive for test conditions N2 and N5.

* FIGURE 15. Crosstalk test circuit for table XIII - Continued.

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* FIGURE 16. Crosstalk test circuit for table XIII.

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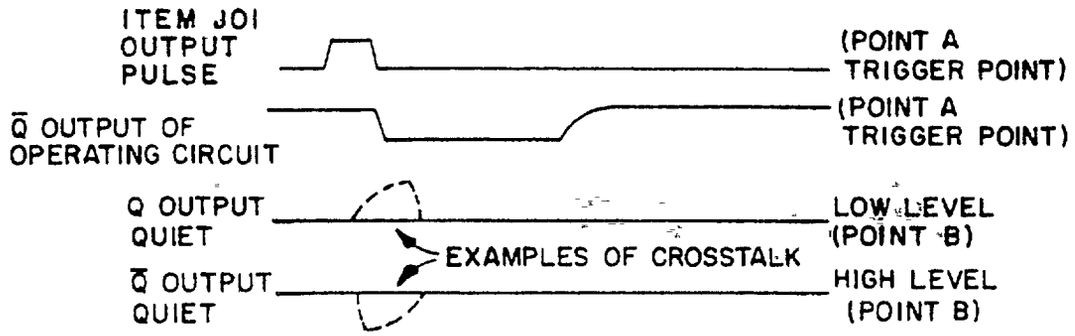
NOTES:
1. R1 = 50Ω ±5%, 1 W.
2. The following capacitances are measured using item C02 at the test fixture pins with the module removed, at a frequency of 1 MHz ±10%, and will include stray wiring capacitance but not oscilloscope probe capacitance. Glass trimmer capacitors may be added to conform to the required values.

   a. C1 = 4 pF maximum.
   b. C2 = 5 ±1 pF.

3. Set the internal trigger slope of item D05 to negative for test conditions N7, N9, N10 and N12 and positive for test conditions N8 and N11.

* FIGURE 16. Crosstalk test circuit for table XIII - Continued.

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NOTES.
1. Adjust the output of item J01 to obtain the following pulse:

   PA = 4 V ±5%.
   PW (2 V level ±0.1 V) = 125 ns ±10%.
   $t_{TLH} = t_{THL}$ = 10 ns ±10%.
   PRP ≥ 200 µs ±10%.

2. To examine the output waveforms under test, item D05 shall be internally triggered using the signal on channel A (probe Y01-1).
3. For each test condition, the maximum pulse width detected at point B, measured at 1.0 V above the low level on the Q output and at 2.0 V below the high level on the Q̄ output, shall not exceed the limits of table XIII.

* FIGURE 17. Crosstalk test circuit waveforms for table XIII.

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* TABLE XIV. Zener applied potential and trigger voltage high test.

Test condition	Parameter	Connect	Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
P1	$\overline{Q1}$ Zener applied potential	5	3.8	5.5	V	18
P2	$\overline{Q2}$ Zener applied potential	33	3.8	5.5	↓	↓
P3	Q1 Trigger voltage high	3	0.0	0.4		
P4	Q2 Trigger voltage high	37	0.0	0.4		

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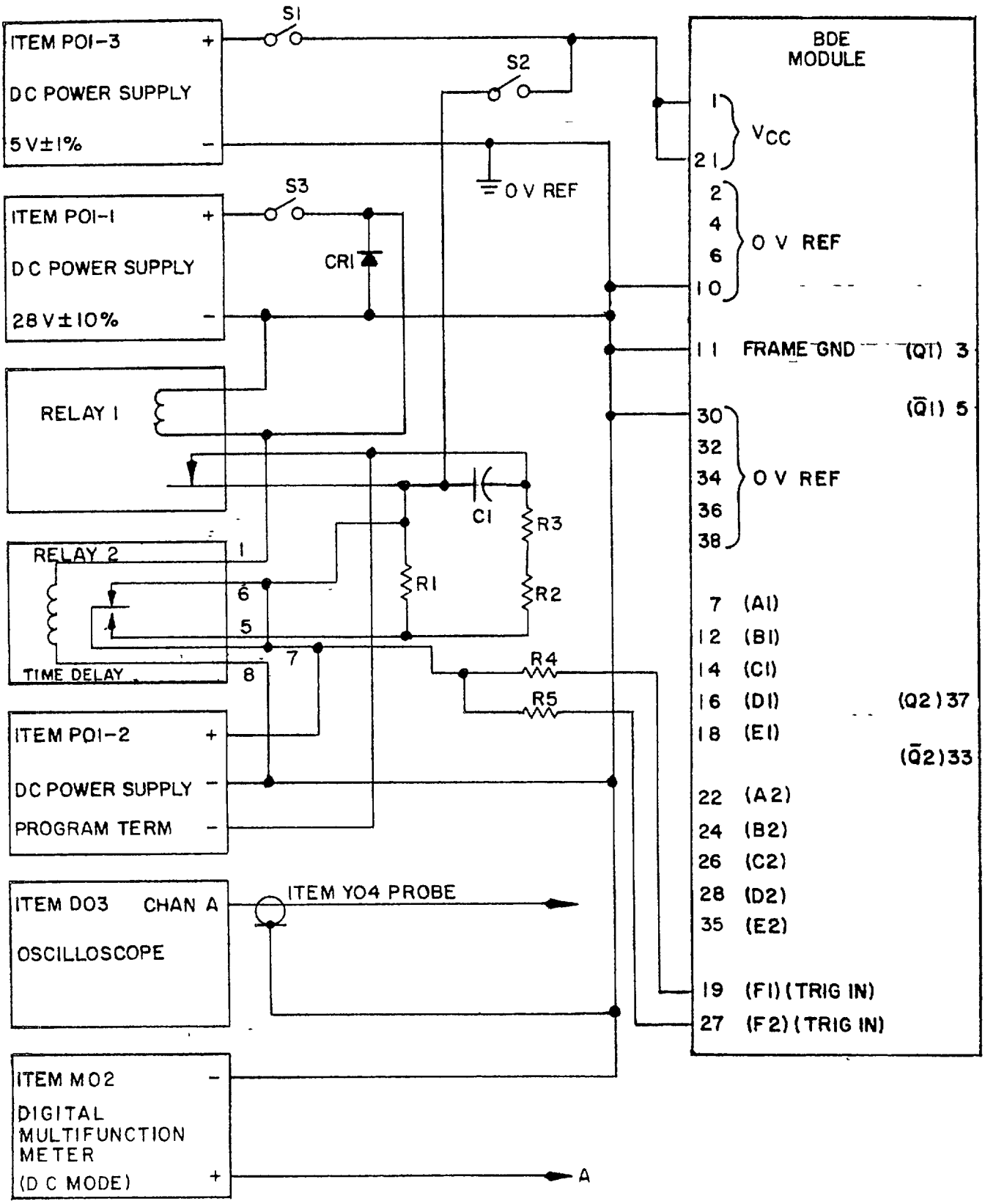
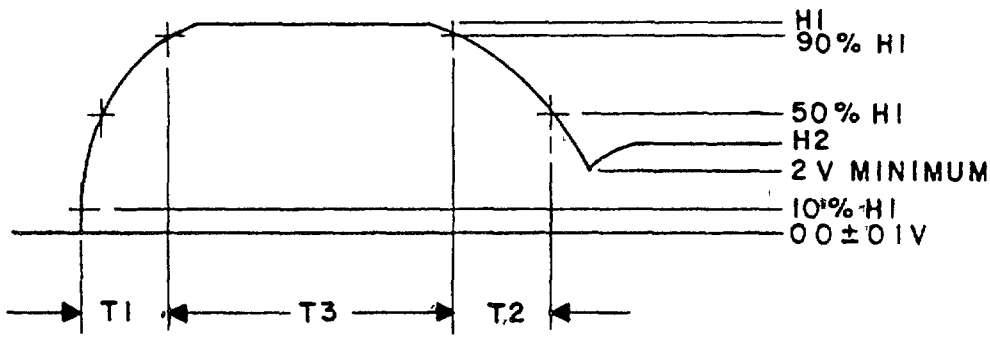


FIGURE 18. Zener applied potential and trigger voltage high test circuit for table XIV.

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NOTES:
1. R1 = 649Ω ±1%, 1/8 W.
2. R2 = 301Ω ±1%, 1/8 W.
3. R3 = 178Ω ±1%, 1/8 W.
4. R4 = R5 = 154Ω ±1%, 1/8 W.
5. C1 = 22 µF ±10%.
6. CR1 = 1N647 or equivalent.
7. Relay 1 = 28 V relay.
8. Relay 2 = Agastat model 2112 or equivalent.
9. S1 = S2 = S3 = SPST switch.
10. Resistors R3, R4, R5, and capacitor C1 are programming components for item P01-2 and are based on a programming constant of 100Ω per V.
11. Using item D03 with item Y04, adjust the time delay relay to obtain the following output waveform at pin 7 of the time delay relay after switch S3 is closed.



Waveform parameters.

HI = 11.25 V ±1%.
H2 = 5.0 V ±10%.

T1 = 100 $^{+100}_{-50}$ ms.

T2 = 100 $^{+400}_{-50}$ ms.

T3 = 0.5 sec ±10%.

12. Item P01-2 should be selected to ensure that the programming terminal is floating (-).
13. Test sequence:

    a. Close switch S3.
    b. Wait 2 seconds minimum.
    c. Make voltage measurements for test conditions.
    d. Open switch S3.

14. For test conditions P1 and P2, switch S1 is open and S2 is closed. For test condition, P3 and P4, switch S1 is closed and S2 is open.

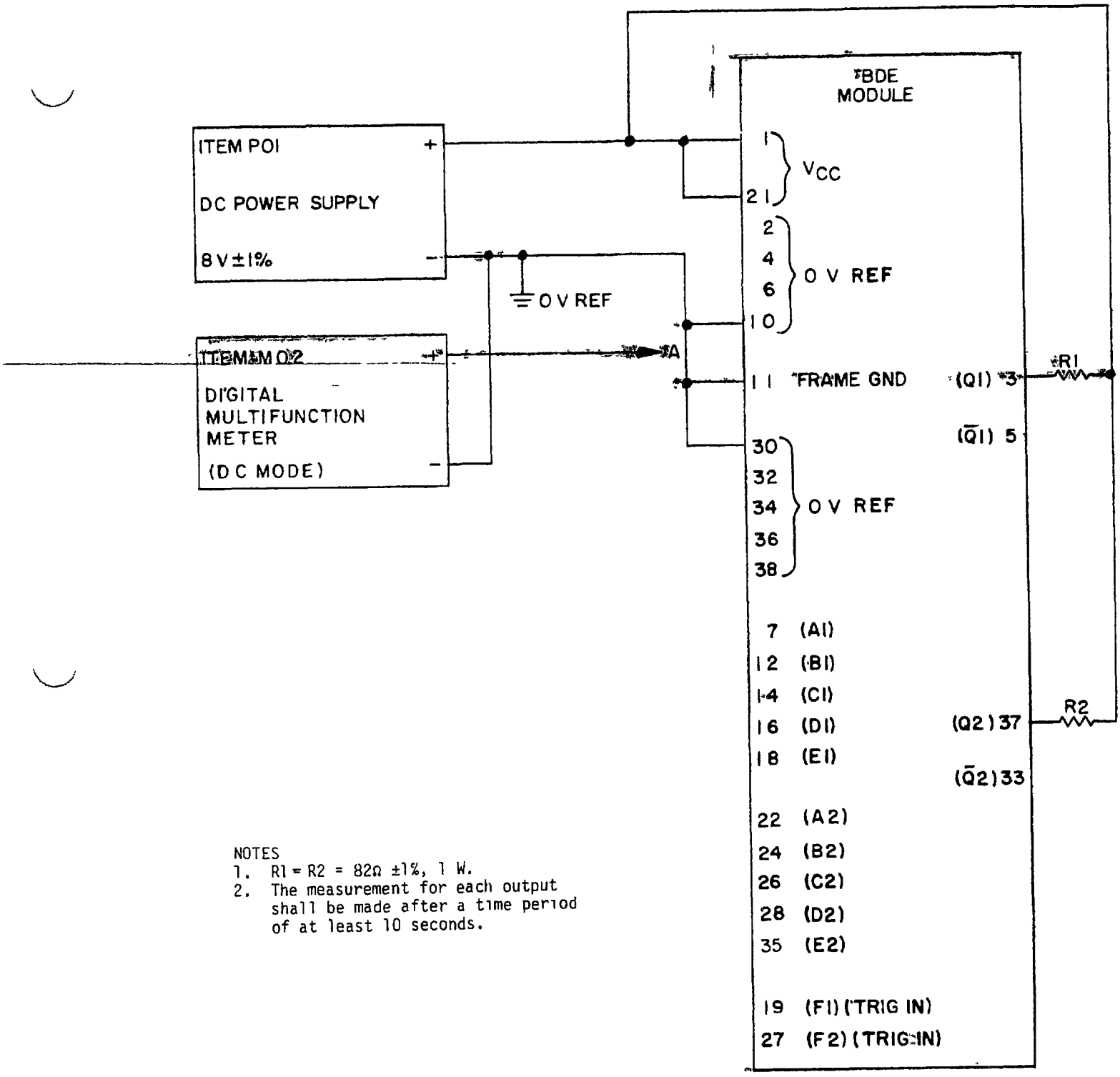
* FIGURE 18. Zener applied potential and trigger voltage high test circuit for table XIV - Continued.

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* TABLE XV. V_{OUT} (low-loaded) test.

Test condition	Parameter	Connect	Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
Q1	Q1 V_{OUT} (low loaded)	3	0.0	1.1	V	19
Q2	Q2 V_{OUT} (low loaded)	37	0.0	1.1	V	19

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NOTES
1. R1 = R2 = 82Ω ±1%, 1 W.
2. The measurement for each output shall be made after a time period of at least 10 seconds.

* FIGURE 19. V_{OUT} (low-loaded) test circuit for table XV.

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